

OSCAT  
Revision History

| Release | Modul name         | Rev | Date     | History                                                                                                                   |
|---------|--------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------|
| 335     | MONTH_TO_STRING    | 1.2 | 16.07.24 | added boundary check to respect array limits                                                                              |
| 334     | CONSTANTS_SETUP    | 1.1 | 03.02.21 | corrected error in decades                                                                                                |
| 334     | CRC_GEN            | 2.1 | 03.02.21 | new improved code by dennis koepke                                                                                        |
| 334     | REVERSE            | 1.2 | 03.02.21 | new improved code by rebler wolfgang                                                                                      |
| 334     | SET_DATE           | 2.4 | 03.02.21 | new better readable code                                                                                                  |
| 334     | CALENDAR_CALC      | 1.7 | 03.02.21 | changed LDT to LOCAL_DT for compatibility with IEC61131-3                                                                 |
| 334     | RTC_2              | 1.4 | 03.02.21 | changed LDT to LOCAL_DT for compatibility with IEC61131-3                                                                 |
| 334     | Calendar Struct    | 1.4 | 03.02.21 | changed LDT, LDATE, LTOD to LOCAL_DT, LOCAL_DATE, LOCAL_TOD                                                               |
| 333     | REAL_TO_STRF       | 1.8 | 02.01.12 | added input parameter D to specify the decimal separator                                                                  |
| 333     | _STRING_TO_BUFFER  | 1.4 | 02.01.12 | functions now returns the next position after the input string                                                            |
| 333     | _BUFFER_INSERT     | 1.5 | 02.01.12 | functions now returns the next position after the input string                                                            |
| 333     | SET_DATE           | 2.3 | 29.12.11 | improved performance                                                                                                      |
| 333     | RTC_2              | 1.4 | 27.04.11 | fixed error with local time calculation                                                                                   |
| 332     | LIST_SORT_L        |     |          | deleted Module                                                                                                            |
| 332     | LIST_RETRIEVE_MIN  |     |          | deleted Module                                                                                                            |
| 332     | LIST_RETRIEVE_MAX  |     |          | deleted Module                                                                                                            |
| 332     | GEN_PULSE          | 1.5 | 08.04.11 | PTL and PTH was exchanged                                                                                                 |
| 332     | CALENDAR_CALC      | 1.6 | 06.04.11 | night was calculated wrong                                                                                                |
| 332     | REAL_TO_FRAC       | 1.1 | 06.04.11 | optimized code                                                                                                            |
| 332     | FILTER_MAV_W       | 1.4 | 26.03.11 | corrected an error formula                                                                                                |
| 332     | FACT               | 1.5 | 26.03.11 | fact is using constants.math                                                                                              |
| 332     | CONSTANTS_MATH     | 1.1 | 26.03.11 | added array facts                                                                                                         |
| 332     | DATE_ADD           | 1.8 | 22.03.11 | corrected error in formula                                                                                                |
| 332     | LIST_RETRIEVE_LAST | 2.0 | 21.03.11 | new Module                                                                                                                |
| 332     | LIST_SORT_L        | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_RETRIEVE_MAX  | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_RETRIEVE_MIN  | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_RETRIEVE      | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_NEXT          | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_LEN           | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_INSERT        | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_GET           | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_CLEAN         | 2.0 | 21.03.11 | every element starts with SEP                                                                                             |
| 332     | LIST_ADD           | 2.0 | 21.03.11 | new module                                                                                                                |
| 332     | MAX3               | 1.1 | 18.03.11 | improved performance                                                                                                      |
| 332     | MIN3               | 1.1 | 18.03.11 | improved performance                                                                                                      |
| 332     | MID3               | 1.2 | 18.03.11 | improved performance                                                                                                      |
| 332     | F_POLY             | 1.1 | 18.03.11 | improved performance                                                                                                      |
| 332     | F_QUAD             | 1.1 | 18.03.11 | improved performance                                                                                                      |
| 332     | ONTIME             | 2.5 | 18.03.11 | reset was not working                                                                                                     |
| 331     | _RMP_B             | 1.2 | 19.02.11 | new enhanced code                                                                                                         |
| 331     | FRMP_B             | 1.0 | 17.01.11 | new module                                                                                                                |
| 331     | CLK_PULSE          | 1.2 | 16.02.11 | corrected an error when timer overflows                                                                                   |
| 331     | GEN_SQ             | 1.3 | 16.02.11 | corrected an error when timer overflows                                                                                   |
| 331     | GEN_PULSE          | 1.4 | 16.02.11 | corrected an error when timer overflows                                                                                   |
| 330     | CALENDAR_CALC      | 1.5 | 02.02.11 | added input H to specify twilight                                                                                         |
| 330     | DATE_ADD           | 1.7 | 02.02.11 | fixed a problem, W was not calculated                                                                                     |
| 330     |                    |     | 02.02.11 | removed building code in external library                                                                                 |
| 321     | LEAP_OF_DATE       | 1.4 | 28.01.11 | improved performance                                                                                                      |
| 321     | DAY_OF_YEAR        | 1.4 | 28.01.11 | improved performance                                                                                                      |
| 321     | DATE_ADD           | 1.6 | 27.01.11 | improved performance                                                                                                      |
| 321     | TIMER_P4           | 1.4 | 26.01.11 | changed code to allow events over midnight                                                                                |
| 321     | TIMER_EVENT        | 1.1 | 26.01.11 | changed type of last to DT                                                                                                |
| 321     | SUN_TIME           | 1.7 | 25.01.11 | using function sun_midday<br>Changed calculation of sun_declination<br>Changed input utc from DT to Date<br>Added input H |
| 321     | SUN_MIDDAY         | 1.0 | 25.01.11 | new module                                                                                                                |
| 321     | FILTER_W           | 1.2 | 25.01.11 | fixed error in formula                                                                                                    |
| 321     | BUFFER_SEARCH      | 1.4 | 19.01.11 | Return -1 if nothing found<br>Ign=true will ignore case                                                                   |
| 321     | DAYS_DELTA         | 1.3 | 25.01.11 | improved performance                                                                                                      |
| 321     | DST                | 1.5 | 24.01.11 | improved performance                                                                                                      |
| 321     | LEAP_DAY           | 1.2 | 24.01.11 | improved performance                                                                                                      |
| 321     | YEAR_END           | 1.1 | 24.01.11 | improved performance                                                                                                      |

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| 321 | BLIND_ACTUATOR     | 1.4  | 08.03.10 | position before control                                                                                 |
| 321 | BLIND_CONTROL      | 1.5  | 08.03.10 | position before control<br>Delta := sene / 2                                                            |
| 321 | HEAT_METER         | 1.10 | 23.01.11 | fixed a problem with pulse_mode                                                                         |
| 321 | FLOW_METER         | 1.0  | 23.01.11 | new module                                                                                              |
| 321 | SET_DATE           | 2.2  | 22.01.11 | improved performance                                                                                    |
| 321 | NETWORK_BUFFER     |      | 21.01.11 | deleted unused data structure                                                                           |
| 321 | URL                |      | 21.01.11 | deleted unused data structure                                                                           |
| 321 | COUNT_SUBSTRING    | 1.0  | 20.01.11 | new module                                                                                              |
| 321 | RTC_2              | 1.3  | 20.01.11 | changed offset from hours to minutes                                                                    |
| 321 | _BUFFER_UPPERCASE  | 1.0  | 20.01.11 | new Module                                                                                              |
| 321 | CHK_REAL           | 1.0  | 20.01.11 | new module                                                                                              |
| 321 | COUNT_DR           | 1.1  | 20.01.11 | corrected init value to 16#FFFFFFF                                                                      |
| 321 | Global_Constants   |      | 20.01.11 | removed nework buffers<br>Added list_length                                                             |
| 321 | REAL_TO_FRAC       | 1.0  | 19.01.11 | new module                                                                                              |
| 321 | GCD                | 1.0  | 19.01.11 | new module                                                                                              |
| 321 | _ARRAY_SORT        | 2.0  | 19.01.11 | new faster algorithm using quicksort                                                                    |
| 321 | DEW_CON            | 1.6  | 19.01.11 | Return 0 if RH <= 0                                                                                     |
| 321 | LIST_SORT_L        | 1.2  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_SORT_L        | 1.2  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_RETRIEVE_MIN  | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_RETRIEVE_MAX  | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_RETRIEVE      | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_NEXT          | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_LEN           | 1.2  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_INSERT        | 1.2  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_GET           | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | LIST_CLEAN         | 1.1  | 19.01.11 | changed string(255) to string(list_length)                                                              |
| 321 | FT_PT1             | 1.11 | 18.01.11 | avoid underrun of out                                                                                   |
| 321 | SCALE_R            | 1.2  | 18.01.11 | avoid division by 0                                                                                     |
| 321 | SCALE_D            | 1.2  | 18.01.11 | avoid division by 0                                                                                     |
| 321 | SCALE_B            | 1.1  | 18.01.11 | avoid division by 0                                                                                     |
| 321 | DEAD_BAND          | 1.2  | 18.01.11 | Assign 0 before return                                                                                  |
| 321 | CONSTANTS_LOCATION | 1.2  | 18.01.11 | added spoken language<br>Deleted holidays for more flexibility                                          |
| 321 | CALENDAR_CALC      | 1.4  | 18.01.11 | added input holidays to specify local holidays<br>Changed call of function sun_time dt >> date          |
| 321 | HOLIDAY            | 2.0  | 18.01.11 | using user specified array for holidays<br>Friday, saturday and sunday is also displayed on output name |
| 321 | EVENTS             | 1.0  | 18.01.11 | New Module                                                                                              |
| 321 | ONTIME             | 2.4  | 17.01.11 | init will not clear ontime and cycles, only rst clears these values                                     |
| 321 | REFLECT            | 1.0  | 16.01.11 | New Module                                                                                              |
| 321 | CRC_CHECK          |      | 16.01.11 | Deleted and replaced by CRC_GEN                                                                         |
| 321 | CRC_GEN            | 2.0  | 16.01.11 | New code and features                                                                                   |
| 321 | Tank_VOL2          | 1.0  | 10.03.10 | New Module                                                                                              |
| 321 | Tank_VOL1          | 1.0  | 10.03.10 | New Module                                                                                              |
| 321 | CIRCLE_SEG         | 1.0  | 10.03.10 | New Module                                                                                              |
| 321 | ACTUATOR_3P        | 2.0  | 29.01.10 | new features and new code                                                                               |
| 321 | CALENDAR_CALC      | 1.3  | 23.01.10 | sun_rise, sun_set and sun_midday are now calculated in local time                                       |
| 321 | CALENDAR           | 1.3  | 23.10.10 | sun_rise, sun_set and sun_midday are now in local time                                                  |
| 321 | BLIND_CONTROL_S    | 1.2  | 23.01.10 | fixed a bug during calibration<br>Added T_UP and T_DN                                                   |
| 321 | ENERGY             | 1.2  | 16.01.10 | avoid the string cal in comments for codesys import bug                                                 |
| 321 | TIMER_1            | 2.2  | 16.12.09 | fixed error over midnight                                                                               |
| 321 | ESR_MON_X8         | 1.2  | 01.12.09 | changed ESR_OUT to be I/O                                                                               |
| 321 | ESR_MON_R4         | 1.4  | 01.12.09 | changed ESR_OUT to be I/O                                                                               |
| 321 | ESR_MON_B8         | 1.3  | 01.12.09 | changed ESR_OUT to be I/O                                                                               |
| 321 | ESR_COLLECT        | 1.4  | 01.12.09 | changes pos to be I/O<br>Changed output array size to 32 elements                                       |
| 320 |                    |      | 24.11.09 | Network Components exported into Network Library                                                        |
| 311 | Global_Constants   |      | 13.11.09 | added network short and long size<br>Default string_length = 250                                        |
| 311 | FT_PIDWL           | 1.3  | 13.11.09 | fixed code for negative KP                                                                              |
| 311 | GEN_PULSE          | 1.3  | 13.11.09 | corrected error                                                                                         |
| 311 | HEAT_METER         | 1.9  | 13.11.09 | C is now calculated correctly                                                                           |
| 311 | LTIME_TO_UTC       | 1.7  | 13.11.09 | corrected an error formula                                                                              |
| 311 | ESR_COLLECT        | 1.3  | 12.11.09 | output pos was not pointing to last value                                                               |

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| 311 | _BUFFER_INSERT    | 1.4 | 12.11.09 | code optimization                                                            |
| 311 | _STRING_TO_BUFFER | 1.3 | 12.11.09 | limit end to size - 1                                                        |
| 311 | BUFFER_COMP       | 1.1 | 12.11.09 | improved performance                                                         |
| 311 | BUFFER_SEARCH     | 1.3 | 12.11.09 | limit end to array size                                                      |
| 311 | BUFFER_TO_STRING  | 1.5 | 12.11.09 | limit start and stop to size of array                                        |
| 311 | BLIND_SHADE_S     | 1.0 | 12.11.09 | New Module                                                                   |
| 311 | BLIND_INPUT       | 1.7 | 12.11.09 | added state 139<br>set defaults for QU, QD, PO                               |
| 311 | GRAY_TO_BYTE      | 1.0 | 09.11.09 | New Module                                                                   |
| 311 | BYTE_TO_GRAY      | 1.0 | 09.11.09 | New Module                                                                   |
| 311 | TREND             | 1.0 | 21.08.09 | new Module                                                                   |
| 311 | SDT_NH3           | 1.0 | 20.08.09 | new Module                                                                   |
| 311 | SDD_NH3           | 1.0 | 20.08.09 | new Module                                                                   |
| 311 | K_TO_C            | 1.1 | 19.08.09 | fixed error in calculation                                                   |
| 311 | C_TO_K            | 1.1 | 19.08.09 | fixed error in calculation                                                   |
| 310 | BLIND_CONTROL_S   | 1.1 | 27.07.09 | major code changes and updates                                               |
| 310 | HEAT_METER        | 1.8 | 27.07.09 | output C is now calculated in J/h                                            |
| 310 | BLIND_CONTROL     | 1.4 | 27.07.09 | changed status management                                                    |
| 310 | BLIND_ACTUATOR    | 1.3 | 27.07.09 | changes status to be 12X                                                     |
| 310 | BLIND_SET         | 1.0 | 27.07.09 | new Module                                                                   |
| 310 | BLIND_INPUT       | 1.6 | 27.07.09 | changed status 0 to 130                                                      |
| 310 | BIN_TO_DWORD      | 1.2 | 26.07.09 | optimized code                                                               |
| 310 | BIN_TO_BYTE       | 1.2 | 26.07.09 | optimized code                                                               |
| 310 | SUN_TIME          | 1.6 | 26.07.09 | fixed a problem with wrong midday calculation                                |
| 310 | STACK_32          | 2.0 | 25.07.09 | changed inputs to allow simultaneous read and write                          |
| 310 | STACK_16          | 2.0 | 25.07.09 | changed inputs to allow simultaneous read and write                          |
| 310 | FIFO_32           | 2.0 | 24.07.09 | changed inputs to allow read and write in one cycle                          |
| 310 | FIFO_16           | 2.0 | 24.07.09 | changed inputs to allow read and write in one cycle                          |
| 310 | TANK_LEVEL        | 1.1 | 24.07.09 | changed parameters of tonof                                                  |
| 310 | TONOF             | 1.2 | 21.07.09 | fixed a timing problem                                                       |
| 310 | TIMECHECK         | 1.0 | 19.07.09 | new Module                                                                   |
| 310 | METER_STAT        | 1.3 | 18.07.09 | changed all outputs to be I/O                                                |
| 310 | DWORD_OF_WORD     | 1.0 | 18.07.09 | new Module                                                                   |
| 310 | WORD_OF_BYTE      | 1.4 | 18.07.09 | added type conversions for compatibility reasons                             |
| 310 | DWORD_OF_BYTE     | 1.3 | 18.07.09 | added type conversions for compatibility reasons                             |
| 310 | GEN_SQ            | 1.2 | 18.07.09 | improved accuracy                                                            |
| 310 | MULTI_IN          | 1.4 | 18.07.09 | improved performance                                                         |
| 310 | _RMP_NEXT         | 1.4 | 18.07.09 | improved performance                                                         |
| 310 | GEN_PULSE         | 1.2 | 18.07.09 | improved performance                                                         |
| 305 | DATE_ADD          | 1.5 | 29.03.09 | improved performance                                                         |
| 305 | DEC_8             | 1.3 | 28.03.09 | replaced multiple assignments                                                |
| 305 | DAYS_IN_YEAR      | 1.0 | 27.03.09 | new Module                                                                   |
| 305 | DAYS_IN_MONTH     | 1.0 | 27.03.09 | new Module                                                                   |
| 305 | MONTH_OF_DATE     | 1.3 | 27.03.09 | new improved code                                                            |
| 305 | BUFFER_TO_STRING  | 1.4 | 23.03.09 | avoid writing to input stop                                                  |
| 305 | _BUFFER_INSERT    | 1.3 | 23.03.09 | avoid writing to input pos                                                   |
| 305 | DRIVER_4C         | 1.0 | 23.03.09 | New Module                                                                   |
| 305 | HEAT_METER        | 1.7 | 23.03.09 | real constants updated to new syntax using dot<br>Corrected error in formula |
| 305 | DELAY             | 1.5 | 23.03.09 | corrected non standard write to input N                                      |
| 305 | DIMM_2            | 1.0 | 22.03.09 | New Module                                                                   |
| 305 | METER             | 1.4 | 18.03.09 | last was not updated<br>Code improvements                                    |
| 305 | FIFO_32           | 1.4 | 14.03.09 | replaced double assignments                                                  |
| 305 | FIFO_16           | 1.4 | 14.03.09 | replaced double assignments                                                  |
| 305 | SCHEDULER         | 1.1 | 14.03.09 | replaced double assignments                                                  |
| 305 | GEN_BIT           | 1.2 | 14.03.09 | replaced double assignments                                                  |
| 305 | CLICK_DEC         | 1.1 | 14.03.09 | replaced double assignments                                                  |
| 305 | STORE_8           | 1.2 | 14.03.09 | replaced double assignments                                                  |
| 305 | LTCH_4            | 1.3 | 14.03.09 | replaced double assignments                                                  |
| 305 | SHR_8UDE          | 1.2 | 14.03.09 | replaced double assignments                                                  |
| 305 | SHR_4UDE          | 1.2 | 14.03.09 | replaced double assignments                                                  |
| 305 | FF_D4E            | 1.3 | 14.03.09 | replaced double assignments                                                  |
| 305 | FF_D2E            | 1.3 | 14.03.09 | replaced double assignments                                                  |
| 305 | TREND_DW          | 1.2 | 14.03.09 | replaced double assignments                                                  |
| 305 | GEN_PW2           | 1.1 | 14.03.09 | replaced double assignments                                                  |
| 305 | MANUAL_1          | 1.2 | 14.03.09 | replaced double assignments                                                  |

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| 305 | INTERLOCK_4    | 1.1  | 14.03.09 | replaced double assignments                                                                      |
| 305 | CONSTANTS_PHYS | 1.1  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | SEQUENCE_8     | 1.5  | 13.03.09 | renamed output any to qx for compatibility reasons                                               |
| 305 | SEQUENCE_4     | 1.5  | 13.03.09 | renamed output any to qx for compatibility reasons                                               |
| 305 | AIR_DENSITY    | 1.3  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | BOILER         | 1.2  | 13.03.09 | real constants updated to new systax using dot<br>replaced double assignments                    |
| 305 | DEW_CON        | 1.5  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | DEW_RH         | 1.2  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | DEW_TEMP       | 1.6  | 13.03.09 | real constants updated to new systax using dot<br>function will now return -273.15 °C for RH = 0 |
| 305 | HEAT_INDEX     | 1.3  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | HEAT_TEMP      | 1.2  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | LEGIONELLA     | 1.3  | 13.03.09 | real constants updated to new systax using dot<br>using new module sequence_8                    |
| 305 | T_AVG24        | 1.2  | 13.03.09 | improved code                                                                                    |
| 305 | TEMP_EXT       | 1.3  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | WATER_CP       | 1.2  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | WATER_DENSITY  | 1.2  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | WATER_ENTHALPY | 1.1  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | WCT            | 1.2  | 13.03.09 | real constants updated to new systax using dot                                                   |
| 305 | BLIND_SHADE    | 1.4  | 13.03.09 | improved code                                                                                    |
| 305 | TUNE2          | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | TUNE           | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | SRAMP          | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | CONTROL_SET1   | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | CONTROL_SET2   | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | DEAD_BAND_A    | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | DEAD_ZONE      | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | DEAD_ZONE2     | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_DERIV       | 1.5  | 11.03.09 | improved code                                                                                    |
| 305 | FT_INT2        | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PID         | 2.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PIDW        | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PIDWL       | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PIWL        | 1.3  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PT1         | 1.10 | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | FT_PT2         | 1.5  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | SPEED          | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | PT_TP_F        | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | PRESSURE       | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | LENGTH         | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | F_TO_PT        | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | F_TO_C         | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | ENERGY         | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | C_TO_F         | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | ASTRO          | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | TC_S           | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | METER_STAT     | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | M_TX           | 1.4  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | CALIBRATE      | 1.3  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | ALARM_2        | 1.1  | 11.03.09 | improved code                                                                                    |
| 305 | TEMP_PT        | 1.7  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | TEMP_NTC       | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | SENSOR_INT     | 1.3  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | RES_SI         | 1.2  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | RES_PT         | 1.3  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | RES_NTC        | 1.1  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | MULTI_IN       | 1.3  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | PWM_PW         | 1.5  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | PWM_DC         | 1.4  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | GEN_SQR        | 1.4  | 11.03.09 | real constants updated to new systax using dot<br>set default amplitude to 1.0                   |
| 305 | GEN_SIN        | 1.6  | 11.03.09 | real constants updated to new systax using dot                                                   |
| 305 | GEN_RMP        | 1.4  | 10.03.09 | real constants updated to new systax using dot                                                   |
| 305 | WORD_TO_RANGE  | 1.1  | 10.03.09 | improved code                                                                                    |
| 305 | STAIR2         | 1.4  | 10.03.09 | real constants updated to new systax using dot                                                   |
| 305 | STAIR          | 1.4  | 10.03.09 | real constants updated to new systax using dot                                                   |

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|     |               |     |          |                                                                                      |
|-----|---------------|-----|----------|--------------------------------------------------------------------------------------|
| 305 | SH_2          | 1.6 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | SEL2_OF_3     | 1.1 | 10.03.09 | improved code                                                                        |
| 305 | MIX           | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | FILTER_WAV    | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | AIN1          | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ARRAY_VAR     | 1.3 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | ARRAY_TREND   | 1.3 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | ARRAY_HAV     | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ARRAY_GAV     | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ARRAY_AVG     | 1.3 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | CTANH         | 1.1 | 10.03.09 | new faster code                                                                      |
| 305 | CTAN          | 1.1 | 10.03.09 | new faster code                                                                      |
| 305 | R2_SET        | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | R2_ABS        | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | IP4_TO_STRING | 1.1 | 10.03.09 | improved code                                                                        |
| 305 | POLYNOM_INT   | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | FT_AVG        | 1.5 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | TRIANGLE_A    | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ELLIPSE_C     | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_ZANG       | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_YANG       | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_XANG       | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_PAR        | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_NUL        | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_NORM       | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | V3_ANG        | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | SUN_TIME      | 1.5 | 10.03.09 | improved code<br>Calculation is done on every call to allow for moving installations |
| 305 | TANH          | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | TANC          | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | SQRTN         | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | SIGN_R        | 1.4 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | SINC          | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | SIGMOID       | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | RND           | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | RDM           | 1.9 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | MODR          | 1.5 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | LANGEVIN      | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | LAMBERT_W     | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | INV           | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | GOLD          | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | GDF           | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | GAUSSCD       | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | GAUSS         | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | GAMMA         | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | FRACT         | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ERFC          | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ERF           | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | DEG           | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | D_TRUNC       | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | COTH          | 1.4 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | COSH          | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | CMP           | 1.1 | 10.03.09 | added type conversion for compatibility reasons                                      |
| 305 | CEIL2         | 1.3 | 10.03.09 | use correct statement real_to_DINT                                                   |
| 305 | ATANH         | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ATAN2         | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ASINH         | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | AGDF          | 1.1 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ACOTH         | 1.2 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | ACOSH         | 1.3 | 10.03.09 | real constants updated to new systax using dot                                       |
| 305 | TEMP_SI       | 1.5 | 10.03.09 | removed nested comments                                                              |
| 305 | TEMP_NI       | 1.5 | 10.03.09 | removed nested comments                                                              |
| 305 | M_T           | 1.4 | 10.03.09 | removed nested comments<br>Removed double ssignments                                 |
| 305 | FT_INT        | 2.2 | 10.03.09 | removed nested comments                                                              |
| 305 | EXPN          | 1.2 | 10.03.09 | removed nested comments<br>Real constants written with dot                           |
| 305 | DAY_OF_MONTH  | 2.1 | 10.03.09 | removed nested comments                                                              |

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|-----|--------------------|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 305 | CRC_GEN            | 1.5 | 10.03.09 | removed nested comments                                                                                                                                                             |
| 305 | IS_IP4             | 1.1 | 10.03.09 | new code                                                                                                                                                                            |
| 305 | IP4_DECODE         | 1.1 | 10.03.09 | new faster algorithm                                                                                                                                                                |
| 305 | CALENDAR_CALC      | 1.2 | 10.03.09 | added calculation of work_week, sun_midday, sun_heighth<br>sun_position will only be calculated once per 25 seconds when SPE = TRUE<br>dst will only become true when dst_en = true |
| 305 | CALENDAR           | 1.2 | 10.03.09 | added work_week, sun_midday, sun_heighth                                                                                                                                            |
| 305 | BLIND_CONTROL_S    | 1.0 | 09.03.09 | new Module                                                                                                                                                                          |
| 305 | _RMP_NEXT          | 1.3 | 09.03.09 | input E was ignored<br>Removed double assignments                                                                                                                                   |
| 305 | DT_SIMU            | 1.2 | 08.03.09 | added increment by cycle mode                                                                                                                                                       |
| 305 | SUN_POS            | 2.1 | 07.03.09 | refraction is added after angle is normalized<br>deleted 10 second lockout<br>added output for astronomical heighth h                                                               |
| 305 | REFRACTION         | 1.1 | 07.03.09 | using new formula                                                                                                                                                                   |
| 304 | TIMER_P4           | 1.3 | 28.02.09 | array was not correctly initialized                                                                                                                                                 |
| 304 | UTC_TO_LTIME       | 1.9 | 27.02.09 | added type conversions to avoid warnings under codesys 3.0                                                                                                                          |
| 304 | SUN_POS            | 2.0 | 23.01.09 | new code with better accuracy                                                                                                                                                       |
| 304 | LTIME_TO_UTC       | 1.6 | 27.02.09 | added type conversions to avoid warnings under codesys 3.0                                                                                                                          |
| 304 | SELECT_8           | 1.1 | 27.02.09 | renamed input en to e                                                                                                                                                               |
| 304 | M_TX               | 1.3 | 27.02.09 | deleted unnecessary init with 0                                                                                                                                                     |
| 304 | M_D                | 1.2 | 27.02.09 | deleted unnecessary init with 0                                                                                                                                                     |
| 304 | REAL_TO_STRF       | 1.7 | 27.02.09 | added a missing zero for IN < 1                                                                                                                                                     |
| 304 | SECOND_TO_TIME     | 1.2 | 24.02.09 | renamed input to IN                                                                                                                                                                 |
| 304 | MINUTE_TO_TIME     | 1.2 | 24.02.09 | renamed input to IN                                                                                                                                                                 |
| 304 | HOUR_TO_TIME       | 1.2 | 24.02.09 | renamed input to IN                                                                                                                                                                 |
| 304 | DAY_TO_TIME        | 1.1 | 24.02.09 | renamed input to IN                                                                                                                                                                 |
| 304 | FILTER_MAV_W       | 1.3 | 23.02.09 | limit N to max array index                                                                                                                                                          |
| 304 | FILTER_MAV_DW      | 1.3 | 23.02.09 | limit N to max array index                                                                                                                                                          |
| 304 | INC1               | 1.2 | 23.02.09 | when inc1 is called with X >= N inc will continue with 0                                                                                                                            |
| 304 | DELAY              | 1.4 | 23.02.09 | corrected an index problem                                                                                                                                                          |
| 304 | _RMP_NEXT          | 1.2 | 20.02.09 | improved algorithm and added T_L                                                                                                                                                    |
| 304 | TREND_DW           | 1.1 | 20.02.09 | added outputs TU, TD and D                                                                                                                                                          |
| 304 | BLIND_SHADE        | 1.3 | 08.02.09 | added calendar data structure<br>Added shading position to be configured                                                                                                            |
| 304 | CALENDAR_CALC      | 1.1 | 08.02.09 | corrected wrong night calculation<br>Added sun position data                                                                                                                        |
| 304 | CALENDAR           | 1.1 | 08.02.09 | added sun position data                                                                                                                                                             |
| 304 | METER              | 1.3 | 08.02.09 | changed MX to be I/O<br>Make sure calculation works for cycle times < 1ms                                                                                                           |
| 304 | HEAT_METER         | 1.6 | 08.02.09 | changed output Y to I/O                                                                                                                                                             |
| 304 | LAMBERT_W          | 1.0 | 08.02.09 | New Module                                                                                                                                                                          |
| 304 | T_AVG24            | 1.1 | 08.02.09 | added outputs T24_MIN and T24_MAX<br>Changed sign of input OFS                                                                                                                      |
| 304 | DIMM_I             | 2.3 | 26.01.09 | dimming will reverse at 0<br>Rst will override set                                                                                                                                  |
| 304 | FT_PIDWL           | 1.1 | 25.01.09 | multiply differential part with KP                                                                                                                                                  |
| 304 | FT_PIWL            | 1.2 | 25.01.09 | now also works with negative K                                                                                                                                                      |
| 304 | SCALE_X8           | 1.2 | 24.01.09 | corrected error in formula                                                                                                                                                          |
| 304 | FADE               | 1.3 | 24.01.09 | deleted unused var FF                                                                                                                                                               |
| 304 | ACTUATOR_2P        | 2.1 | 24.01.09 | deleted unused var inb                                                                                                                                                              |
| 304 | _RMP_NEXT          | 1.1 | 24.01.09 | deletes unused vars tmp1 and tmp2                                                                                                                                                   |
| 304 | GEO_TO_DEG         | 1.0 | 22.01.09 | New Module                                                                                                                                                                          |
| 304 | BLIND_INPUT        | 1.5 | 22.01.09 | added auto mode for IN<br>Added pulse mode for D1 and D2                                                                                                                            |
| 304 | TO_UPPER           | 1.3 | 16.01.09 | corrected an error in module                                                                                                                                                        |
| 303 | TEMP_NTC           | 1.0 | 30.12.08 | New Module                                                                                                                                                                          |
| 303 | RES_NTC            | 1.0 | 30.12.08 | New Module                                                                                                                                                                          |
| 303 | CLICK              | 2.0 | 28.12.08 | new code using library modules                                                                                                                                                      |
| 303 | CONSTANTS_LOCATION | 1.1 | 23.12.08 | added french holidays                                                                                                                                                               |
| 303 | CONSTANTS_LANGUAGE | 1.2 | 23.12.08 | added french language                                                                                                                                                               |
| 303 | TIMER_2            | 1.5 | 22.12.08 | make sure timer_2 is executed in the first cycle                                                                                                                                    |
| 303 | URL_DECODE         | 1.1 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |
| 303 | TO_UML             | 1.1 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |
| 303 | LIST_SORT_L        | 1.1 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |
| 303 | LIST_INSERT        | 1.1 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |
| 303 | HTML_DECODE        | 1.2 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |
| 303 | FILL               | 1.1 | 17.12.08 | changed function CHR to CHR_TO_STRING                                                                                                                                               |

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|-----|------------------|-----|----------|-------------------------------------------------------------------------------|
| 303 | CHARNAME         | 1.4 | 17.12.08 | changed function CHR to CHR_TO_STRING                                         |
| 303 | CHR_TO_STRING    | 1.3 | 17.12.08 | changed name of function chr to chr_to_string                                 |
| 303 | SUN_POS          | 1.6 | 17.12.08 | angles below horizon are displayed in negative degrees                        |
| 303 | ONTIME           | 2.3 | 17.12.08 | fixed an error when in would be true for more then 49 days                    |
| 303 | M_T              | 1.3 | 17.12.08 | code optimization                                                             |
| 303 | TP_X             | 1.3 | 17.12.08 | code optimization                                                             |
| 303 | FADE             | 1.2 | 17.12.08 | FADE input F was inverted                                                     |
| 303 | F_LIN2           | 1.1 | 17.12.08 | optimized formula                                                             |
| 303 | _RMP_NEXT        | 1.0 | 24.11.08 | New Module                                                                    |
| 303 | FILTER_MAV_W     | 1.2 | 24.11.08 | added type conversions for compatibility reasons<br>avoid divide by 0 f N = 0 |
| 303 | FILTER_MAV_DW    | 1.2 | 24.11.08 | added type conversions for compatibility reasons<br>avoid divide by 0 f N = 0 |
| 303 | ACTUATOR_2P      | 2.0 | 23.11.08 | new code using library modules                                                |
| 303 | GEN_PULSE        | 1.1 | 23.11.08 | added default enq = TRUE                                                      |
| 303 | MANUAL           | 1.0 | 21.11.08 | New Module                                                                    |
| 303 | BAND_B           | 1.0 | 21.11.08 | New Module                                                                    |
| 303 | TREND_DW         | 1.0 | 21.11.08 | New Module                                                                    |
| 303 | AUTORUN          | 1.0 | 20.11.08 | New Module                                                                    |
| 303 | _RMP_W           | 1.1 | 20.11.08 | set default for E to be TRUE<br>added Init code                               |
| 303 | _RMP_B           | 1.1 | 20.11.08 | set default for E to be TRUE<br>added Init code                               |
| 303 | SDT_TO_TOD       | 1.1 | 16.11.08 | chaged typecasts to avoid warnings                                            |
| 303 | ONTIME           | 2.2 | 16.11.08 | chaged typecasts to avoid warnings                                            |
| 303 | DT_SIMU          | 1.1 | 16.11.08 | added type conversions for compatibility reasons                              |
| 303 | T_PLC_MS         | 1.2 | 16.11.08 | initialized constants with 0 for compatibility reasons                        |
| 303 | T_PLC_US         | 1.2 | 16.11.08 | initialized constants with 0 for compatibility reasons                        |
| 303 | GLOBAL CONSTANTS |     | 16.11.08 | moved structured constants to global vars for compatibility reasons           |
| 303 | DIMM_I           | 2.2 | 16.11.08 | added Reset Out feature                                                       |
| 302 | BUFFER_COMP      | 1.0 | 14.11.08 | New Module                                                                    |
| 302 | DIMM_I           | 2.1 | 14.11.08 | added soft_dimm feature                                                       |
| 302 | CLICK_MODE       | 1.0 | 13.11.08 | New Module                                                                    |
| 302 | SCALE_R          | 1.1 | 13.11.08 | corrected formula for negative gradient                                       |
| 302 | SCALE_D          | 1.1 | 13.11.08 | corrected formula for negative gradient                                       |
| 302 | ONTIME           | 2.1 | 10.11.08 | increased internal resolution to milliseconds                                 |
| 302 | FILTER_I         | 1.0 | 08.11.08 | New Module                                                                    |
| 302 | ACTUATOR_A       | 1.0 | 08.11.08 | New Module                                                                    |
| 302 | TIMER_EXT        | 1.1 | 06.11.08 | added default = true for ena input                                            |
| 302 | FT_DERIV         | 1.4 | 06.11.08 | optimized code for better performance                                         |
| 302 | TIMER_1          | 2.1 | 06.11.08 | set defaults for inputs day and e, improved performance                       |
| 302 | HEAT_METER       | 1.5 | 06.11.08 | changed to FT_INT2 to avoid overrun at hig values                             |
| 302 | FT_PIDW          | 1.1 | 05.11.08 | rewritten code using integrate                                                |
| 302 | FT_INT2          | 1.1 | 05.11.08 | rewritten code using integrate                                                |
| 302 | CTRL_OUT         | 1.1 | 05.11.08 | optimized code for better performance                                         |
| 302 | FT_INT           | 2.1 | 03.11.08 | optimized code using integrate                                                |
| 302 | D_TRUNC          | 1.2 | 03.11.08 | deleted unused variable check                                                 |
| 302 | INTEGRATE        | 1.0 | 03.11.08 | New Module                                                                    |
| 302 | FT_PT2           | 1.4 | 03.11.08 | optimized code and fixed a problem with init                                  |
| 302 | FILTER_DW        | 1.1 | 03.11.08 | fixed overflow problem in formula                                             |
| 302 | FILTER_W         | 1.1 | 03.11.08 | fixed overflow problem in formula                                             |
| 302 | CYCLE_4          | 1.0 | 03.11.08 | New Module                                                                    |
| 302 | SCALE_B8         | 1.4 | 03.11.08 | used wrong factor in formula                                                  |
| 302 | SCALE_B4         | 1.4 | 03.11.08 | used wrong factor in formula                                                  |
| 302 | SCALE_B2         | 1.4 | 03.11.08 | used wrong factor in formula                                                  |
| 302 | T_AVG24          | 1.0 | 03.11.08 | New Module                                                                    |
| 301 | TEMP_PT          | 1.6 | 31.10.08 | optimized code for better performance                                         |
| 301 | _BUFFER_INIT     | 1.2 | 31.10.08 | corrected an error while routine would write outside of arrays                |
| 301 | _BUFFER_CLEAR    | 1.2 | 31.10.08 | corrected an error while routine would write outside of arrays                |
| 301 | D_TRUNC          | 1.1 | 31.10.08 | optimized code for better performance                                         |
| 301 | MODR             | 1.4 | 31.10.08 | changed algorithm to the more common version using floor instead of TRUNC     |
| 301 | PIN_CODE         | 1.0 | 30.10.08 | New Module                                                                    |
| 301 | BYTE_OF_DWORD    | 1.2 | 30.10.08 | optimized code for better performance                                         |
| 301 | WORD_OF_DWORD    | 1.2 | 30.10.08 | optimized code for better performance                                         |
| 301 | TOGGLE           | 1.1 | 30.10.08 | deleted unnecessary init with 0                                               |
| 301 | STORE_8          | 1.1 | 30.10.08 | optimized code for better performance                                         |
| 301 | LTCH_4           | 1.2 | 30.10.08 | deleted unnecessary init with 0                                               |
| 301 | LTCH             | 1.2 | 30.10.08 | deleted unnecessary init with 0                                               |

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|-----|--------------------|-----|----------|------------------------------------------------------------------------------------------------|
| 301 | FF_JKE             | 1.2 | 30.10.08 | optimized code for better performance                                                          |
| 301 | FF_DRE             | 1.2 | 30.10.08 | optimized code for better performance                                                          |
| 301 | FF_D4E             | 1.2 | 30.10.08 | deleted unnecessary init with 0                                                                |
| 301 | FF_D2E             | 1.2 | 30.10.08 | deleted unnecessary init with 0                                                                |
| 301 | MESSAGE_4R         | 1.1 | 27.10.08 | changed _INC to INC1                                                                           |
| 301 | FILTER_WAV         | 1.1 | 27.10.08 | changed _DEC and _INC to DEC1 and INC1                                                         |
| 301 | FILTER_MAV_W       | 1.1 | 27.10.08 | added typecast to avoid warnings                                                               |
| 301 | FILTER_MAV_DW      | 1.1 | 27.10.08 | added typecast to avoid warnings                                                               |
| 301 | FIFO_32            | 1.3 | 25.10.08 | optimized code for better performance                                                          |
| 301 | FIFO_16            | 1.3 | 25.10.08 | optimized code for better performance                                                          |
| 301 | STACK_32           | 1.3 | 25.10.08 | optimized code for better performance                                                          |
| 301 | STACK_16           | 1.3 | 25.10.08 | optimized code for better performance                                                          |
| 301 | ESR_COLLECT        | 1.1 | 27.10.08 | optimized code for better performance                                                          |
| 301 | DELAY              | 1.3 | 27.10.08 | performance improvement                                                                        |
| 301 | INC1               | 1.1 | 27.10.08 | New Module, replacing _INC1                                                                    |
| 301 | DEC1               | 1.1 | 27.10.08 | New Module, replacing _DEC1                                                                    |
| 301 | SIGN_I             | 1.3 | 27.10.08 | optimized code for better performance                                                          |
| 301 | GAUSS              | 1.1 | 27.10.08 | optimized code for better performance                                                          |
| 301 | FT_PIWL            | 1.1 | 27.10.08 | integrator will not be adjusted when ki = 0                                                    |
| 301 | FACT               | 1.4 | 27.10.08 | optimized code for better performance                                                          |
| 301 | CAUCHYCD           | 1.0 | 26.10.08 | New Module                                                                                     |
| 301 | CAUCHY             | 1.0 | 26.10.08 | New Module                                                                                     |
| 301 | NEGX               | 1.0 | 26.10.08 | New Module                                                                                     |
| 301 | INV                | 1.0 | 26.10.08 | New Module                                                                                     |
| 301 | BETA               | 1.0 | 26.10.08 | New Module                                                                                     |
| 301 | GAMMA              | 1.2 | 26.10.08 | optimized code for better performance<br>Improved accuracy for small values by a factor > 1000 |
| 301 | CONSTANTS_MATH     | 1.0 | 26.10.08 | Added 1 / pi and 1 / e                                                                         |
| 301 | MATRIX             | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | DEC_8              | 1.2 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | SCALE_B8           | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | SCALE_B4           | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | SCALE_B2           | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | STAIR2             | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | STAIR              | 1.3 | 26.10.08 | changed input bit to val for compatibility reasons                                             |
| 301 | SCALE_X2           | 1.1 | 26.10.08 | optimized code for better performance                                                          |
| 301 | RND                | 1.1 | 26.10.08 | optimized code for better performance                                                          |
| 301 | RMP_SOFT           | 3.0 | 26.10.08 | New code using _RMP_B                                                                          |
| 301 | GEN_RMP            | 1.3 | 26.10.08 | optimized code for better performance                                                          |
| 301 | TO_UPPER           | 1.2 | 25.10.08 | optimized code for better performance                                                          |
| 301 | TO_LOWER           | 1.2 | 25.10.08 | optimized code for better performance                                                          |
| 301 | CLK_PRG            | 1.3 | 25.10.08 | optimized code for better performance                                                          |
| 301 | SHR8_UDE           | 1.1 | 25.10.08 | optimized code for better performance                                                          |
| 301 | SHR4_UDE           | 1.1 | 25.10.08 | optimized code for better performance                                                          |
| 301 | SHR_4E             | 1.1 | 25.10.08 | optimized code for better performance                                                          |
| 301 | ROUND              | 1.5 | 25.10.08 | optimized code for better performance using constant decades                                   |
| 301 | A_TRIG             | 1.1 | 25.10.08 | optimized code for better performance<br>A_TRIG now works for res = 0                          |
| 301 | RND                | 1.1 | 25.10.08 | optimized code for better performance                                                          |
| 301 | FSTRING_TO_MONTH   | 1.2 | 25.10.08 | optimized code for better performance                                                          |
| 301 | TIMER_EVENT_DECODE | 1.1 | 25.10.08 | using new language defaults                                                                    |
| 301 | FSTRING_TO_WEEK    | 1.1 | 25.10.08 | using new language defaults                                                                    |
| 301 | FSTRING_TO_WEEKDAY | 1.2 | 25.10.08 | using new language defaults                                                                    |
| 301 | WORK_WEEK          | 1.5 | 25.10.08 | optimized code for better performance                                                          |
| 301 | DAY_OF_MONTH       | 2.0 | 25.10.08 | new code using setup constants                                                                 |
| 301 | SET_DATE           | 2.0 | 25.10.08 | new code using setup constants                                                                 |
| 301 | CONSTANTS_SETUP    | 1.0 | 25.10.08 | New Module                                                                                     |
| 301 | ERF                | 1.2 | 25.10.08 | optimized code for better performance                                                          |
| 301 | BINOM              | 1.0 | 25.10.08 | New Module                                                                                     |
| 301 | CHARNAME           | 1.3 | 24.10.08 | optimized code for better performance                                                          |
| 301 | CHARCODE           | 1.2 | 24.10.08 | optimized code for better performance                                                          |
| 301 | INTERLOCK_4        | 1.0 | 23.10.08 | New Module                                                                                     |
| 301 | TANC               | 1.0 | 23.10.08 | New Module                                                                                     |
| 301 | CALENDAR_CALC      | 1.0 | 23.10.08 | New Module                                                                                     |
| 301 | LEGIONELLA         | 1.2 | 23.10.08 | changed code to use new version of timer_1                                                     |
| 301 | TIMER_1            | 2.0 | 23.10.08 | New code, enable input new features (DAYS)                                                     |
| 301 | RMP_W              | 2.0 | 23.10.08 | New code using _RMP_W                                                                          |
| 301 | RMP_B              | 2.0 | 23.10.08 | New code using _RMP_B                                                                          |

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| 301 | _RMP_W             | 1.0 | 23.10.08 | New Module                                                                                                                                      |
| 301 | FLOAT_TO_REAL      | 1.2 | 23.10.08 | fixed a problem with character 8 and 9 behind the comma                                                                                         |
| 301 | LIST_SORT_L        | 1.0 | 23.10.08 | New Module                                                                                                                                      |
| 301 | LIST_RETRIEVE_MIN  | 1.0 | 23.10.08 | New Module                                                                                                                                      |
| 301 | ESR_MON_B8         | 1.2 | 22.10.08 | optimized code for better performance                                                                                                           |
| 301 | DIR_TO_DEG         | 1.1 | 22.10.08 | using language constants                                                                                                                        |
| 301 | CONSTANTS_LANGUAGE | 1.1 | 22.10.08 | New Module                                                                                                                                      |
| 301 | DEG_TO_DIR         | 1.1 | 22.10.08 | using language constants                                                                                                                        |
| 301 | LIST_RETRIEVE_MAX  | 1.0 | 22.10.08 | New Module                                                                                                                                      |
| 301 | DIMM_I             | 2.0 | 22.10.08 | changed structure and code for better performance<br>New features                                                                               |
| 301 | _RMP_B             | 1.0 | 22.10.08 | New Module                                                                                                                                      |
| 301 | F_LAMP             | 1.2 | 22.10.08 | changed structure and code for better performance and external variables                                                                        |
| 301 | EXP_N              | 1.1 | 22.10.08 | optimized code for better performance                                                                                                           |
| 301 | SW_RECONFIG        | 1.0 | 22.10.08 | New Module                                                                                                                                      |
| 301 | CTRL_PWM           | 1.1 | 21.10.08 | optimized code for better performance                                                                                                           |
| 301 | HEAT_METER         | 1.4 | 21.10.08 | optimized code for better performance                                                                                                           |
| 301 | ACTUATOR_2P        | 1.6 | 21.10.08 | optimized code for better performance                                                                                                           |
| 301 | HOLIDAY            | 1.6 | 21.10.08 | using location constants                                                                                                                        |
| 301 | FIND_CHAR          | 1.3 | 21.10.08 | changed setup constants                                                                                                                         |
| 301 | FINDB_NONUM        | 1.3 | 21.10.08 | optimized code for better performance                                                                                                           |
| 301 | FIND_NONUM         | 1.3 | 21.10.08 | optimized code for better performance                                                                                                           |
| 301 | ACTUATOR_PUMP      |     | 21.10.08 | changed code to use ontime rev 2.0                                                                                                              |
| 301 | BURNER             | 1.6 | 21.10.08 | changed code to use ontime rev 2.0                                                                                                              |
| 301 | ONTIME             | 2.0 | 21.10.08 | changed structure and code for better performance and external variables                                                                        |
| 301 | CALENDAR           | 1.0 | 21.10.08 | New Module                                                                                                                                      |
| 301 | LTIME_TO_UTC       | 1.5 | 20.10.08 | changed time_zone_offset from Real to INT, now in Minutes                                                                                       |
| 301 | LOCAL_TIME         |     | 20.10.08 | deleted from library, can be replaced by UTC_TO_LTIME                                                                                           |
| 301 | UTC_TO_LTIME       | 1.8 | 20.10.08 | changes type of input TIME_ZONE_OFFSET from real to int, now is in +/-minutes<br>Deleted outputs DSE and WEEKDAY<br>Changed from FB to Function |
| 301 | CONSTANTS_LOCATION | 1.0 | 20.10.08 | New Module                                                                                                                                      |
| 301 | GLOBAL CONSTANTS   |     | 20.10.08 | changed to structured constants                                                                                                                 |
| 301 | TIMER_2            | 1.4 | 20.10.08 | performance improvement<br>Deleted unnecessary trunc                                                                                            |
| 301 | TO_UPPER           | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | TO_LOWER           | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | ISC_UPPER          | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | ISC_LOWER          | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | ISC_ALPHA          | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | CHARNAME           | 1.2 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | CHARCODE           | 1.1 | 19.10.08 | changed setup constants                                                                                                                         |
| 301 | FSTRING_TO_MONTH   | 1.1 | 19.10.08 | changed language setup constants                                                                                                                |
| 301 | MONTH_TO_STRING    | 1.1 | 19.10.08 | changed language setup constants                                                                                                                |
| 301 | WEEKDAY_TO_STRING  | 1.1 | 19.10.08 | changed language setup constants                                                                                                                |
| 301 | DT_TO_STRF         | 1.1 | 19.10.08 | changed language setup constants                                                                                                                |
| 301 | NEG                |     | 19.10.08 | deleted unnecessary function because -X is faster                                                                                               |
| 301 | PWM_PW             | 1.4 | 19.10.08 | changed type TP_R to TP_X because of name change                                                                                                |
| 301 | PWM_DC             | 1.3 | 19.10.08 | changed type TP_R to TP_X because of name change<br>Improved performance                                                                        |
| 301 | TP_X               | 1.2 | 19.10.08 | renamed module from TP_R for compatibility reasons                                                                                              |
| 301 | IRTRANS_DECODE     | 1.1 | 19.10.08 | added typecast to avoid warnings<br>changed output device to dev for compatibility reasons                                                      |
| 301 | IRTRANS_8          | 1.2 | 19.10.08 | changed input device_code to dev_code and device to dev for compatibility reasons                                                               |
| 301 | IRTRANS_4          | 1.2 | 19.10.08 | changed input device_code to dev_code and device to dev for compatibility reasons                                                               |
| 301 | IRTRANS_1          | 1.2 | 19.10.08 | changed input device_code to dev_code and device to dev for compatibility reasons                                                               |
| 301 | BLIND_ACTUATOR     | 1.2 | 18.10.08 | changed calls for rmp_B because of change in RMP_B                                                                                              |
| 301 | FT_AVG             | 1.4 | 18.10.08 | renamed inout EN to E for compatibility reasons                                                                                                 |
| 301 | BIT_LOAD_DW2       | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | BIT_LOAD_W2        | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | BIT_LOAD_B2        | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | SH_T               | 1.1 | 18.10.08 | renamed inout EN to E for compatibility reasons                                                                                                 |
| 301 | RMP_W              | 1.8 | 18.10.08 | renamed inout EN to E for compatibility reasons                                                                                                 |
| 301 | BIT_TOGGLE_DW      | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | BIT_TOGGLE_W       | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | BIT_TOGGLE_B       | 1.0 | 18.10.08 | New Module                                                                                                                                      |
| 301 | DEW_TEMP           | 1.5 | 18.10.08 | changed to use phys constants                                                                                                                   |
| 301 | DEW_CON            | 1.4 | 18.10.08 | changed to use phys constants                                                                                                                   |

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| 301 | AIR_DENSITY    | 1.2 | 18.10.08 | changed to use phys constants                                                                                                                                     |
| 301 | SUN_POS        | 1.5 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | RDMDW          | 1.2 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | RDM2           | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | RDM            | 1.8 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | RAD            | 1.2 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | OM_TO_F        | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | GEN_SIN        | 1.5 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | GDF            | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | GAMMA          | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | F_TO_OM        | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | ELLIPSE_C      | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | ELLIPSE_A      | 1.1 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | VEKTOR_3       | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | URL            | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | TIMER_EVENT    | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | SDT            | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | REAL2          | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | NETWORK_BUFFER | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | HOLIDAY_DATA   | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | ESR_DATA       | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | CONSTANTS_PHYS | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | COMPLEX        | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | ATAN2          | 1.0 | 18.10.08 | changed to use math constants                                                                                                                                     |
| 301 | SDT_TO_DATE    | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | SDT_TO_DT      | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | SDT_TO_TOD     | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | DT2_TO_SDT     | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | DT_TO_SDT      | 1.0 | 18.10.08 | New Module                                                                                                                                                        |
| 301 | TIMER_P4       | 1.2 | 18.10.08 | added typecast to avoid warnings<br>Changes type of input ofs from int to byte                                                                                    |
| 301 | RMP_B          | 1.8 | 18.10.08 | added typecast to avoid warnings<br>changed input en to e for compatibility reasons                                                                               |
| 301 | FADE           | 1.1 | 18.10.08 | performance improvement<br>changed calls for rmp_w because rmp_w has chaged                                                                                       |
| 277 | MANUAL_1       | 1.1 | 17.10.08 | deleted unnecessary variable m_edge                                                                                                                               |
| 277 | IRTRANS_8      | 1.1 | 17.10.08 | performance improvement                                                                                                                                           |
| 277 | IRTRANS_4      | 1.1 | 17.10.08 | performance improvement                                                                                                                                           |
| 277 | ACTUATOR_UD    | 1.3 | 17.10.08 | deleted unnecessary initialization with 0                                                                                                                         |
| 277 | STACK_16       | 1.2 | 17.10.08 | deleted unnecessary init with 0                                                                                                                                   |
| 277 | STACK_32       | 1.2 | 17.10.08 | deleted unnecessary init with 0                                                                                                                                   |
| 277 | FIFO_32        | 1.2 | 17.10.08 | performance improvement                                                                                                                                           |
| 277 | FIFO_16        | 1.2 | 17.10.08 | performance improvement                                                                                                                                           |
| 277 | BOILER         | 1.1 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | MUX_4          | 1.2 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | MUX_2          | 1.2 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | MUX_R2         | 1.1 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | MUX_R4         | 1.1 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | LIST_LEN       | 1.1 | 16.10.08 | fixed a problem where list_len would only work up to STRING_LENGTH                                                                                                |
| 277 | RMP_W          | 1.8 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | RMP_B          | 1.7 | 16.10.08 | performance improvement                                                                                                                                           |
| 277 | GEN_BIT        | 1.1 | 15.10.08 | performance improvement                                                                                                                                           |
| 277 | MULTI_IN       | 1.2 | 14.10.08 | corrected an error for in_3 overrange detection<br>Improved performance                                                                                           |
| 277 | DEW_CON        | 1.3 | 14.10.08 | performance improvement                                                                                                                                           |
| 277 | UTC_TO_LTIME   | 1.7 | 14.10.08 | Renamed module from LTIME to UTC_TO_LTIME<br>Changed input Time_zone_offset from int to real to allow half hour offset<br>changed function weekday to day_of_week |
| 277 | SET_DATE       | 1.7 | 14.10.08 | optimized code for better performance                                                                                                                             |
| 277 | LTIME_TO_UTC   | 1.4 | 14.10.08 | changes code for efficiency<br>Changes TIME_Zone_Offset from Int to REAL to allow half hour offset                                                                |
| 277 | FILTER_WAV     | 1.0 | 14.10.08 | New Module                                                                                                                                                        |
| 277 | ACTUATOR_PUMP  | 1.3 | 13.10.08 | auto activation can now be disabled when run_every = T#0s                                                                                                         |
| 277 | _DEC1          | 1.0 | 13.10.08 | New Module                                                                                                                                                        |
| 277 | _INC1          | 1.0 | 13.10.08 | New Module                                                                                                                                                        |
| 277 | FILTER_MAV_W   | 1.0 | 13.10.08 | New Module                                                                                                                                                        |
| 277 | FILTER_MAV_DW  | 1.0 | 13.10.08 | New Module                                                                                                                                                        |
| 277 | FT_AVG         | 1.3 | 10.10.08 | improved performance                                                                                                                                              |

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| 277 | TIMER_P4           | 1.1  | 09.10.08 | changed function year to year_of_date<br>changed function weekday to day_of_week<br>changed structure of program to be more efficient<br>Added set and reset of channel at specific times<br>Changed input DIS to ENQ |
| 277 | FILTER_W           | 1.0  | 09.10.08 | New Module                                                                                                                                                                                                            |
| 277 | FILTER_DW          | 1.0  | 09.10.08 | New Module                                                                                                                                                                                                            |
| 277 | MESSAGE_4R         | 1.0  | 08.10.08 | New Module                                                                                                                                                                                                            |
| 277 | TIMER_EXT          | 1.0  | 07.10.08 | New Module                                                                                                                                                                                                            |
| 277 | TIMER_1            | 1.7  | 07.10.08 | changed function weekday to day_of_week                                                                                                                                                                               |
| 277 | DCF_77             | 1.10 | 07.10.08 | changed function weekday to day_of_week                                                                                                                                                                               |
| 277 | DAY_OF_WEEK        | 1.4  | 07.10.08 | changed name of function from weekday to day_of_week                                                                                                                                                                  |
| 277 | WORK_WEEK          | 1.4  | 07.10.08 | changed function year to year_of_date<br>changed function weekday to day_of_week                                                                                                                                      |
| 277 | TIMER_2            | 1.3  | 07.10.08 | chaged function month to month_of_date<br>changed function weekday to day_of_week                                                                                                                                     |
| 277 | TEMP_EXT           | 1.2  | 07.10.08 | chaged function month to month_of_date                                                                                                                                                                                |
| 277 | MONTH_END          | 1.1  | 07.10.08 | changed function year to year_of_date<br>changed function month to month_of_date                                                                                                                                      |
| 277 | METER_STAT         | 1.1  | 07.10.08 | changed function year to year_of_date<br>changed function month to month_of_date<br>changed function weekday to day_of_week                                                                                           |
| 277 | LEAP_DAY           | 1.1  | 07.10.08 | changed function month to month_of_date                                                                                                                                                                               |
| 277 | HOLIDAY            | 1.5  | 07.10.08 | changed function year to year_of_date<br>changed function weekday to day_of_week<br>changed function month to month_of_date                                                                                           |
| 277 | DT_TO_STRF         | 1.0  | 07.10.08 | New Module                                                                                                                                                                                                            |
| 277 | DST                | 1.4  | 07.10.08 | changed function year to year_of_date                                                                                                                                                                                 |
| 277 | DAY_OF_YEAR        | 1.3  | 07.10.08 | changed name of function year to year_of_date                                                                                                                                                                         |
| 277 | DATE_ADD           | 1.4  | 07.10.08 | changed function year to year_of_date<br>changed function month to month_of_date                                                                                                                                      |
| 277 | MONTH_OF_DATE      | 1.2  | 07.10.08 | changed name of function month to month_of_date                                                                                                                                                                       |
| 277 | YEAR_OF_DATE       | 1.4  | 07.10.08 | changed name of function year to year_of_date                                                                                                                                                                         |
| 277 | FLOAT_TO_REAL      | 1.1  | 02.10.08 | fixed an error, characters 8 and 9 would not be converted                                                                                                                                                             |
| 277 | DIMM_I             | 1.4  | 30.09.08 | added input val to supply value for set operation<br>Added setup parameter INIT_VAL                                                                                                                                   |
| 277 | DEC_TO_DWORD       | 1.1  | 30.09.08 | corrected an error where decoding of characters 8 and 9 would fail                                                                                                                                                    |
| 277 | DEC_TO_INT         | 1.1  | 30.09.08 | changed length of input dtring from 20 to 10<br>corrected an error where decoding of characters 8 and 9 would fail<br>corrected an error where decoding of characters 8 and 9 would fail                              |
| 277 | DEC_TO_BYTE        | 1.1  | 30.09.08 | changed length of input dtring from 20 to 10<br>corrected an error where decoding of characters 8 and 9 would fail                                                                                                    |
| 277 | SCHEDULER_2        | 1.0  | 29.09.08 | New Module                                                                                                                                                                                                            |
| 277 | SCHEDULER          | 1.0  | 28.09.08 | New Module                                                                                                                                                                                                            |
| 277 | HEAT_TEMP          | 1.1  | 26.09.08 | moved T_INT from config to Input<br>Added config variable H<br>Corrected error in formula                                                                                                                             |
| 277 | GEN_PW2            | 1.0  | 26.09.08 | New Module                                                                                                                                                                                                            |
| 277 | BLIND_INPUT        | 1.4  | 26.09.08 | corrected an error while PO and AO was set during automatic modes                                                                                                                                                     |
| 277 | FSTRING_TO_MONTH   | 1.0  | 25.09.08 | New Module                                                                                                                                                                                                            |
| 277 | FSTRING_TO_DT      | 1.0  | 24.09.08 | New Module                                                                                                                                                                                                            |
| 277 | MANUAL_2           | 1.0  | 22.09.08 | New Module                                                                                                                                                                                                            |
| 277 | DEBOUNCE           | 1.0  | 22.09.08 | New Module                                                                                                                                                                                                            |
| 277 | WEEKDAY_TO_STRING  | 1.0  | 21.09.08 | New Module                                                                                                                                                                                                            |
| 277 | MONTH_TO_STRING    | 1.0  | 21.09.08 | New Module                                                                                                                                                                                                            |
| 277 | GLOBAL CONSTANTS   |      | 21.09.08 | added new section Language Setup<br>added new section country setup                                                                                                                                                   |
| 277 | OCT_TO_BYTE        | 1.1  | 20.09.08 | changed length of input dtring from 20 to 10                                                                                                                                                                          |
| 277 | HEX_TO_BYTE        | 1.1  | 20.09.08 | changed length of input dtring from 20 to 5                                                                                                                                                                           |
| 277 | FSTRING_TO_DWORD   | 1.1  | 20.09.08 | changed length of input dtring from 20 to 40                                                                                                                                                                          |
| 277 | FSTRING_TO_BYTE    | 1.1  | 20.09.08 | changed length of input dtring from 20 to 12                                                                                                                                                                          |
| 277 | BIN_TO_DWORD       | 1.1  | 20.09.08 | changed length of input dtring from 20 to 40                                                                                                                                                                          |
| 277 | BIN_TO_BYTE        | 1.1  | 20.09.08 | changed length of input dtring from 20 to 12                                                                                                                                                                          |
| 277 | DT_SIMU            | 1.0  | 11.09.08 | New Module                                                                                                                                                                                                            |
| 276 | TMIN               | 1.0  | 21.06.08 | New Module                                                                                                                                                                                                            |
| 276 | TMAX               | 1.0  | 20.06.08 | New Module                                                                                                                                                                                                            |
| 276 | FSTRING_TO_WEEKDAY | 1.1  | 18.07.08 | changed nested call of left(trim()) for compatibility reasons                                                                                                                                                         |
| 276 | TOF_1              | 1.0  | 17.07.08 | New Module                                                                                                                                                                                                            |

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| 276 | CLICK_DEC          | 1.0 | 17.07.08 | New Module                                                                                                                                                              |
| 276 | CLICK_CNT          | 1.0 | 16.07.08 | New Module                                                                                                                                                              |
| 276 | A_TRIG             | 1.0 | 16.07.08 | New Module                                                                                                                                                              |
| 276 | BLIND_INPUT        | 1.3 | 16.07.08 | rewritten code to allow for more functionality<br>added single click mode with single_switch<br>added Debounce for inputs S1 and S2<br>added double_click functionality |
| 276 | FF_RSE             | 1.0 | 16.07.08 | New Module                                                                                                                                                              |
| 276 | PULSE_T            | 1.0 | 15.07.08 | New Module                                                                                                                                                              |
| 276 | JD2000             | 1.0 | 15.07.08 | New Module                                                                                                                                                              |
| 276 | REFRACTION         | 1.0 | 14.07.08 | New Module                                                                                                                                                              |
| 275 | STAIR2             | 1.2 | 30.06.08 | added type conversions to avoid warnings under codesys 3.0                                                                                                              |
| 275 | CEIL2              | 1.2 | 30.06.08 | added type conversions to avoid warnings under codesys 3.0                                                                                                              |
| 275 | ERF                | 1.1 | 30.06.08 | added type conversions to avoid warnings under codesys 3.0                                                                                                              |
| 275 | SUN_POS            | 1.4 | 30.06.08 | added type conversions to avoid warnings under codesys 3                                                                                                                |
| 275 | CTRL_PID           | 2.0 | 30.06.08 | renamed Module FT_PID to CTRL_PID<br>rewritten code to use modular approach                                                                                             |
| 275 | CTRL_PI            | 2.0 | 30.06.08 | renamed Module FT_PI to CTRL_PI<br>rewritten code to use modular approach                                                                                               |
| 275 | INT_TO_BCDC        | 1.1 | 30.06.08 | changed name INT_TO_BCD to INT_TO_BCDC to avoid collision with util.lib<br>corrected error in code                                                                      |
| 275 | BCDC_TO_INT        | 1.1 | 30.06.08 | changed name BCD_TO_INT to BCDC_TO_INT to avoid collision with util.lib                                                                                                 |
| 275 | GLOBAL CONSTANTS   |     | 30.06.08 | reduced size of string constants to 253 to avoid problems with target aixia DCUF                                                                                        |
| 275 | SEQUENCE_64        | 1.0 | 29.06.08 | New Module                                                                                                                                                              |
| 275 | INC2               | 1.0 | 29.06.08 | New Module                                                                                                                                                              |
| 275 | GEN_PULSE          | 1.0 | 29.06.08 | New Module                                                                                                                                                              |
| 275 | FLOW_CONTROL       | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | TP_1               | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | TP_1D              | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | LIST_INSERT        | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | LIST_CLEAN         | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | LIST_RETRIEVE      | 1.0 | 28.06.08 | New Module                                                                                                                                                              |
| 275 | LIST_NEXT          | 1.0 | 25.06.08 | New Module                                                                                                                                                              |
| 275 | LIST_LEN           | 1.0 | 25.06.08 | New Module                                                                                                                                                              |
| 275 | FLOAT_TO_REAL      | 1.0 | 22.06.08 | New Module                                                                                                                                                              |
| 275 | DWORD_TO_STRB      | 1.3 | 20.06.08 | improved algorithm for higher speed                                                                                                                                     |
| 275 | BYTE_TO_STRB       | 1.3 | 20.06.08 | improved algorithm for higher speed                                                                                                                                     |
| 275 | IRTRANS_1          | 1.1 | 20.06.08 | improved algorithm for higher speed                                                                                                                                     |
| 275 | LIST_PARSE         | 1.0 | 20.06.08 | New Module                                                                                                                                                              |
| 275 | DEC_TO_DWORD       | 1.0 | 20.06.08 | New Module                                                                                                                                                              |
| 275 | DEC_TO_INT         | 1.0 | 20.06.08 | New Module                                                                                                                                                              |
| 275 | DEC_TO_BYTE        | 1.0 | 20.06.08 | New Module                                                                                                                                                              |
| 275 | FSTRING_TO_WEEK    | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | FSTRING_TO_WEEKDAY | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | TIMER_EVENT_DECODE | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | FSTRING_TO_DWORD   | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | FSTRING_TO_BYTE    | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | BIN_TO_DWORD       | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | BIN_TO_BYTE        | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | OCTAL_TO_BYTE      | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | OCTAL_TO_DWORD     | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | HEX_TO_DWORD       | 1.4 | 18.06.08 | changed input hex to STRING(20)<br>function now ignores wrong characters                                                                                                |
| 275 | HEX_TO_BYTE        | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | DEL_CHARS          | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | CLEAN              | 1.0 | 18.06.08 | New Module                                                                                                                                                              |
| 275 | TIMER_P4           | 1.0 | 17.06.08 | New Module                                                                                                                                                              |
| 275 | MANUAL_4           | 1.0 | 17.06.08 | New Module                                                                                                                                                              |
| 275 | MANUAL_1           | 1.0 | 17.06.08 | New Module                                                                                                                                                              |
| 275 | HTML_DECODE        | 1.1 | 16.06.08 | changed nested call of replace for better compatibility                                                                                                                 |
| 275 | CHARNAME           | 1.1 | 16.06.08 | changed nested call of concat for better compatibility                                                                                                                  |
| 275 | LEAP_DAY           | 1.0 | 15.06.08 | New Module                                                                                                                                                              |
| 275 | MONTH_BEGIN        | 1.0 | 15.06.08 | New Module                                                                                                                                                              |
| 275 | MONTH_END          | 1.0 | 15.06.08 | New Module                                                                                                                                                              |
| 275 | YEAR_END           | 1.0 | 15.06.08 | New Module                                                                                                                                                              |
| 275 | FT_PT1             | 1.9 | 14.06.08 | improved algorithm for higher accuracy                                                                                                                                  |
| 275 | SIGN_R             | 1.3 | 14.06.08 | improved algorithm for higher speed                                                                                                                                     |
| 275 | DEAD_ZONE          | 1.1 | 14.06.08 | improved algorithm for higher speed                                                                                                                                     |

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| 275 | DEAD_BAND         | 1.1 | 14.06.08 | improved algorithm for higher speed                                                   |
| 275 | DEAD_BAND_A       | 1.0 | 14.06.08 | New Module                                                                            |
| 275 | SH_2              | 1.5 | 14.06.08 | set default for input N = 16                                                          |
| 275 | FT_AVG            | 1.2 | 14.06.08 | set default for input EN = TRUE and N = 32                                            |
| 275 | FT_PIDWL          | 1.0 | 13.06.08 | New Module                                                                            |
| 275 | FT_PIWL           | 1.0 | 13.06.08 | New Module                                                                            |
| 275 | MS_TO_BFT         | 1.0 | 12.06.08 | New Module                                                                            |
| 275 | BFT_TO_MS         | 1.0 | 12.06.08 | New Module                                                                            |
| 275 | BUFFER_TO_STRING  | 1.3 | 12.06.08 | check for pointer overrun<br>change type of inputs start and stop to UINT             |
| 275 | COUNT_DR          | 1.0 | 12.06.08 | New Module                                                                            |
| 275 | RTC_2             | 1.2 | 12.06.08 | improved performance                                                                  |
| 275 | DIR_TO_DEG        | 1.0 | 11.06.08 | New Module                                                                            |
| 275 | DEG_TO_DIR        | 1.0 | 11.06.08 | New Module                                                                            |
| 275 | BIT_OF_DWORD      | 1.2 | 06.06.08 | changes type of input N from byte to int                                              |
| 275 | HOURL_OF_DT       | 1.0 | 06.06.08 | New Module                                                                            |
| 275 | MINUTE_OF_DT      | 1.0 | 06.06.08 | New Module                                                                            |
| 275 | SECOND_OF_DT      | 1.0 | 06.06.08 | New Module                                                                            |
| 275 | FT_PIDW           | 1.0 | 03.06.08 | New Module                                                                            |
| 275 | FT_PIW            | 1.0 | 03.06.08 | New Module                                                                            |
| 275 | FT_PDT1           | 1.0 | 03.06.08 | New Module                                                                            |
| 275 | FT_PD             | 1.0 | 03.06.08 | New Module                                                                            |
| 275 | CTRL_PWM          | 1.0 | 03.06.08 | New Module                                                                            |
| 275 | FT_PID            | 2.0 | 03.06.08 | New Module                                                                            |
| 275 | FT_PI             | 2.0 | 03.06.08 | New Module                                                                            |
| 275 | CTRL_OUT          | 1.0 | 02.06.08 | New Module                                                                            |
| 275 | CTRL_IN           | 1.0 | 02.06.08 | New Module                                                                            |
| 275 | R2_SET            | 1.0 | 02.06.08 | New Module                                                                            |
| 275 | FT_INT2           | 1.0 | 02.06.08 | New Module                                                                            |
| 275 | FT_INT            | 2.0 | 02.06.08 | rewritten with clear code<br>added output LIM<br>switched back to 32 Bit resolution   |
| 275 | HYST_2            | 1.2 | 02.06.08 | improved algorithm for higher speed                                                   |
| 275 | HYST_1            | 1.1 | 02.06.08 | improved algorithm for higher speed                                                   |
| 275 | HYST              | 1.0 | 02.06.08 | New Module                                                                            |
| 275 | TANK_LEVEL        | 1.0 | 01.06.08 | New Module                                                                            |
| 275 | ACTUATOR_COIL     | 1.0 | 01.06.08 | New Module                                                                            |
| 274 | F_POLY            | 1.0 | 20.05.08 | New Module                                                                            |
| 274 | SEL2_OF_3B        | 1.0 | 19.05.08 | New Module                                                                            |
| 274 | ALARM_2           | 1.0 | 19.05.08 | New Module                                                                            |
| 274 | RDM               | 1.7 | 18.05.08 | changed constant E to E1                                                              |
| 274 | SEL2_OF_3         | 1.0 | 18.05.08 | New Module                                                                            |
| 274 | RDMDW             | 1.1 | 18.05.08 | changed constant E to E1                                                              |
| 274 | SCALE_R           | 1.0 | 18.05.08 | New Module                                                                            |
| 274 | SCALE_D           | 1.0 | 18.05.08 | New Module                                                                            |
| 274 | SCALE_B           | 1.0 | 18.05.08 | New Module                                                                            |
| 274 | GLOBAL CONSTANTS  |     | 18.05.08 | added charname lists<br>renamed E to E1                                               |
| 274 | SCALE             | 1.0 | 18.05.08 | New Module                                                                            |
| 274 | REPLACE_CHARS     | 1.0 | 14.05.08 | New Module                                                                            |
| 274 | CHARNAME          | 1.0 | 14.05.08 | New Module                                                                            |
| 274 | CHARCODE          | 1.0 | 14.05.08 | New Module                                                                            |
| 274 | BUFFER_TO_STRING  | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH |
| 274 | BUFFER_SEARCH     | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH |
| 274 | _STRING_TO_BUFFER | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH |
| 274 | _BUFFER_INSERT    | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH |
| 274 | HTML_DECODE       | 1.0 | 03.05.08 | New Module                                                                            |
| 274 | HTML_ENCODE       | 1.0 | 03.05.08 | New Module                                                                            |
| 274 | GOLD              | 1.0 | 27.04.08 | New Module                                                                            |
| 274 | AGDF              | 1.0 | 27.04.08 | New Module                                                                            |
| 274 | GDF               | 1.0 | 27.04.08 | New Module                                                                            |
| 274 | GLOBAL CONSTANTS  |     | 27.04.08 | added new constant PI05 and PI025                                                     |
| 274 | PERIOD2           | 1.0 | 27.04.08 | New Module                                                                            |
| 274 | URL_DECODE        | 1.0 | 27.04.08 | New Module                                                                            |
| 274 | URL_ENCODE        | 1.0 | 27.04.08 | New Module                                                                            |

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| 274 | IS_URLCHR        | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274 | WORD_OF_BYTE     | 1.3 | 23.04.08 | reverse order of inputs to be more logical                                                                                                                             |
| 274 | DWORD_OF_BYTE    | 1.2 | 23.04.08 | reverse order of inputs to be more logical                                                                                                                             |
| 274 | IP4_TO_STRING    | 1.0 | 22.04.08 | New Module                                                                                                                                                             |
| 274 | AIN1             | 1.2 | 22.04.08 | corrected error in formula when code_min was set<br>corrected error when sign bit was used<br>optimized code for better performance<br>new functionality overflow_code |
| 274 | CLOG             | 1.1 | 20.04.08 | use ATAN2 instead of ATAN                                                                                                                                              |
| 274 | CARG             | 1.1 | 20.04.08 | use ATAN2 instead of ATAN                                                                                                                                              |
| 274 | ATAN2            | 1.0 | 20.04.08 | New Module                                                                                                                                                             |
| 274 | GLOBAL CONSTANTS |     | 19.04.08 | added Setup Parameter NETWORK_BUFFER_SIZE                                                                                                                              |
| 274 | DCF77            | 1.9 | 19.04.08 | added input dsi to allow to set daylight savings time when SDT is TRUE<br>added asynchronous SET input                                                                 |
| 274 | URL_TO_STRING    | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274 | STRING_TO_URL    | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274 | REAL_TO_DW       | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274 | DW_TO_REAL       | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274 | GAUSSCD          | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274 | ERFC             | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274 | ERF              | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274 | GLOBAL CONSTANTS |     | 07.04.08 | deleted unused setup parameter STEP7                                                                                                                                   |
| 274 | YEAR_BEGIN       | 1.2 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274 | YEAR             | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274 | WEEKDAY          | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274 | SET_DATE         | 1.6 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274 | HOLIDAY          | 1.4 | 07.04.08 | improved algorithm for higher speed                                                                                                                                    |
| 274 | EASTER           | 1.3 | 07.04.08 | improved algorithm for higher speed                                                                                                                                    |
| 274 | DAY_OF_DATE      | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274 | GAUSS            | 1.0 | 06.04.08 | New Module                                                                                                                                                             |
| 274 | MODR             | 1.3 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |
| 274 | IS_SORTED        | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |
| 274 | FLOOR2           | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |
| 274 | CEIL2            | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |
| 274 | ARRAY_HAV        | 1.0 | 02.04.08 | New Module                                                                                                                                                             |
| 274 | ARRAY_GAV        | 1.0 | 02.04.08 | New Module                                                                                                                                                             |
| 274 | _ARRAY_MUL       | 1.0 | 02.04.08 | New Module                                                                                                                                                             |
| 274 | _ARRAY_ADD       | 1.0 | 02.04.08 | New Module                                                                                                                                                             |
| 274 | _ARRAY_ABS       | 1.0 | 02.04.08 | New Module                                                                                                                                                             |
| 273 | _ARRAY_SHUFFLE   | 1.2 | 30.03.08 | changed last in rdm2 from pos to i+pos for more randomness                                                                                                             |
| 273 | TO_UML           | 1.0 | 29.03.08 | New Module                                                                                                                                                             |
| 273 | UPPERCASE        | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | TRIM1            | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | TRIM             | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | TICKER           | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | REPLACE_UML      | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)<br>new code to avoid pointer out of range<br>use new function to_uml                                                           |
| 273 | REPLACE_ALL      | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | REAL_TO_STRF     | 1.5 | 29.03.08 | changed STRING to STRING(20)                                                                                                                                           |
| 273 | MIRROR           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | MESSAGE          | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | LOWERCASE        | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_UPPER         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_NUM           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_NCC           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_LOWER         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_HEX           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_CTRL          | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_CC            | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_ALPHA         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | IS_ALNUM         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | HEX_TO_DWORD     | 1.3 | 29.03.08 | changed STRING to STRING(8)                                                                                                                                            |
| 273 | FINDP            | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | FINDB_NUM        | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | FINDB_NONUM      | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | FINDB            | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |
| 273 | FIND_NUM         | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                                                                |

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| 273 | FIND_NONUM        | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | FIND_CTRL         | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | FIND_CHAR         | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | EXEC              | 1.4 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | DWORD_TO_STRH     | 1.3 | 29.03.08 | changed STRING to STRING(8)                                                                                             |
| 273 | DWORD_TO_STRF     | 1.1 | 29.03.08 | changed STRING to STRING(20)<br>limit the output string to max 20 digits                                                |
| 273 | DWORD_TO_STRB     | 1.2 | 29.03.08 | changed STRING to STRING(32)                                                                                            |
| 273 | COUNT_CHAR        | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | CODE              | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | CHR               | 1.2 | 29.03.08 | changed STRING to STRING(1)                                                                                             |
| 273 | CAPITALIZE        | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                                 |
| 273 | BYTE_TO_STRH      | 1.3 | 29.03.08 | changed STRING to STRING(2)                                                                                             |
| 273 | BYTE_TO_STRB      | 1.2 | 29.03.08 | changed STRING to STRING(8)                                                                                             |
| 273 | GLOBAL CONSTANTS  |     | 28.03.08 | Added setup parameter STRING_LENGTH                                                                                     |
| 273 | IS_SORTED         | 1.0 | 28.03.08 | New Module                                                                                                              |
| 273 | TRIME             | 1.0 | 28.03.08 | New Module                                                                                                              |
| 273 | FIX               | 1.0 | 28.03.08 | New Module                                                                                                              |
| 273 | FILL              | 1.0 | 28.03.08 | New Module                                                                                                              |
| 273 | FIND_CTRL         | 1.1 | 26.03.08 | character 127 is now recognized as control                                                                              |
| 273 | FIND_CHAR         | 1.1 | 26.03.08 | char will now accept extended ascii                                                                                     |
| 273 | FIB               | 1.3 | 26.03.08 | function now returns -1 for input < 0 or > 46                                                                           |
| 273 | METER             | 1.2 | 24.03.08 | use data_type real2 to extend accuracy to 15 digits total<br>do not use ft_int which adds unnecessary overhead          |
| 273 | PERIOD            | 1.3 | 22.03.08 | function would deliver wrong results when d1 and dx are a leap_year                                                     |
| 273 | DAYS_DELTA        | 1.2 | 22.03.08 | changed output from int to dint because the total date range is 49710 days                                              |
| 273 | DAY_OF_DATE       | 1.2 | 22.03.08 | changed output from int to dint because the total date range is 49710 days                                              |
| 273 | DATE_ADD          | 1.3 | 22.03.08 | fixed some bugs when month was negative                                                                                 |
| 273 | R2_ABS            | 1.0 | 21.03.08 | New Module                                                                                                              |
| 273 | FRACT             | 1.2 | 21.03.08 | Replaced TRUNC with D_TRUNC for compatibility reasons                                                                   |
| 273 | MODR              | 1.2 | 21.03.08 | Replaced TRUNC with D_TRUNC for compatibility reasons                                                                   |
| 273 | FLOOR2            | 1.0 | 21.03.08 | New Module                                                                                                              |
| 273 | CEIL              | 1.1 | 21.03.08 | Replaced TRUNC with real_to_int for compatibility reasons                                                               |
| 273 | EVEN              | 1.1 | 21.03.08 | Changed type of input IN from int to dint                                                                               |
| 273 | FLOOR             | 1.1 | 21.03.08 | Replaced TRUNC with real_to_int for compatibility reasons                                                               |
| 273 | CEIL2             | 1.0 | 21.03.08 | New Module                                                                                                              |
| 273 | D_TRUNC           | 1.0 | 21.03.08 | New Module                                                                                                              |
| 273 | R2_ADD2           | 1.0 | 20.03.08 | New Module                                                                                                              |
| 273 | R2_MUL            | 1.0 | 20.03.08 | New Module                                                                                                              |
| 273 | R2_ADD            | 1.0 | 20.03.08 | New Module                                                                                                              |
| 273 | TRIM1             | 1.1 | 20.03.08 | avoid to call replace with pos = 0 because some systems will produce an error                                           |
| 273 | TRIM              | 1.1 | 20.03.08 | avoid to call replace with pos = 0 because some systems will produce an error                                           |
| 273 | REAL_TO_STRF      | 1.4 | 20.03.08 | changed trunc to dint because trunc was generating wrong values on wago 842<br>corrected an error with negative numbers |
| 273 | EXEC              | 1.3 | 20.03.08 | Make sure output of exec will always be in the format x.y                                                               |
| 273 | IS_NCC            | 1.0 | 19.03.08 | New Module                                                                                                              |
| 273 | IS_CC             | 1.0 | 19.03.08 | New Module                                                                                                              |
| 273 | CRC_CHECK         | 1.1 | 16.03.08 | changed type of input size to uint<br>improved code for better performance                                              |
| 273 | CRC_GEN           | 1.4 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | PARSET            | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | HEAT_METER        | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | _STRING_TO_BUFFER | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | BUFFER_SEARCH     | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | BUFFER_TO_STRING  | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | _BUFFER_INSERT    | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | GEN_SIN           | 1.4 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | GEN_RDT           | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | FT_PT1            | 1.8 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | FT_INT            | 1.9 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                               |
| 273 | DST               | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>code improvement for better performance                    |
| 273 | ESR_MON_R4        | 1.3 | 16.03.08 | deleted wrong conversion real_to_dword                                                                                  |
| 273 | ARRAY_VAR         | 1.2 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | ARRAY_TREND       | 1.2 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | ARRAY_SUM         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | ARRAY_SPR         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |
| 273 | ARRAY_SDV         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                      |

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| 273 | ARRAY_MIN      | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ONTIME         | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | ARRAY_MAX      | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_AVG      | 1.2 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_SORT    | 1.6 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_SHUFFLE | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_MEDIAN  | 1.5 | 16.03.08 | changed type of input size to uint<br>improved performance                                                                   |
| 273 | _ARRAY_INIT    | 1.3 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | SET_DATE       | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | SET_DT         | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | SUN_POS        | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TC_S           | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TUNE           | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TUNE2          | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>improved performance                                            |
| 273 | WORK_WEEK      | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | DCF77          | 1.8 | 16.03.08 | Changed output weekday to wday and dst to ds for compatibility reasons                                                       |
| 273 | BIT_LOAD_DW    | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | BIT_LOAD_W     | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | BIT_LOAD_B     | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | AIN1           | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | AIN            | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | _BUFFER_CLEAR  | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>changed type of input size to uint<br>deleted unused variable i |
| 273 | _BUFFER_INIT   | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>changed type of input size to uint                              |
| 273 | BYTE_TO_BITS   | 1.1 | 16.03.08 | renamed from byte_to_bit to byte_to_bits for compatibility reasons                                                           |
| 273 | BYTE_TO_BIT    | 1.1 | 16.03.08 | renamed from byte_to_bit to byte_to_bits for compatibility reasons                                                           |
| 273 | SET_TOD        | 1.5 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                                                     |
| 273 | RDM            | 1.6 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                                                     |
| 273 | ROUND          | 1.4 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                                                     |
| 273 | ASINH          | 1.2 | 16.03.08 | changed sequence of calculations to improve performance                                                                      |
| 273 | ACOSH          | 1.2 | 16.03.08 | changed sequence of calculations to improve performance                                                                      |
| 273 | DIFFER         | 1.1 | 16.03.08 | Improved code for higher speed                                                                                               |
| 272 | RDMDW          | 1.0 | 14.03.08 | New Module                                                                                                                   |
| 272 | MULTIME        | 1.1 | 14.03.08 | rounded the result after the last digit                                                                                      |
| 272 | HOUR_TO_TOD    | 1.1 | 14.03.08 | rounded the input after the last digit                                                                                       |
| 272 | HOUR_TO_TIME   | 1.1 | 14.03.08 | rounded the input after the last digit                                                                                       |
| 272 | MINUTE_TO_TIME | 1.1 | 14.03.08 | rounded the input after the last digit                                                                                       |
| 272 | SECOND_TO_TIME | 1.1 | 14.03.08 | rounded the input after the last digit                                                                                       |
| 272 | SET_TOD        | 1.4 | 14.03.08 | changed code to avoid rounding problem at last digit of millisecond                                                          |
| 272 | TC_S           | 1.0 | 13.03.08 | New Module                                                                                                                   |
| 272 | TC_MS          | 1.0 | 13.03.08 | New Module                                                                                                                   |
| 272 | TC_US          | 1.0 | 13.03.08 | New Module                                                                                                                   |
| 272 | SRAMP          | 1.0 | 13.03.08 | New Module                                                                                                                   |
| 272 | FT_INT         | 1.8 | 13.03.08 | changed preset value out_min to -1e37<br>stop integrator at the limits<br>optimized code                                     |
| 272 | CMP            | 1.0 | 12.03.08 | New Module                                                                                                                   |
| 272 | SINH           | 1.3 | 11.03.08 | improved accuracy around 0                                                                                                   |
| 272 | SINC           | 1.0 | 11.03.08 | New Module                                                                                                                   |
| 272 | LANGEVIN       | 1.1 | 11.03.08 | changed formula to avoid problems when x = 0                                                                                 |
| 272 | SQRTN          | 1.2 | 11.03.08 | added result 0 for compatibility reasons                                                                                     |
| 272 | FRACT          | 1.1 | 11.03.08 | added dint_to_real for compatibility reasons<br>now returns 0 for number > 2e9<br>changed input to x                         |
| 272 | RND            | 1.0 | 11.03.08 | New Module                                                                                                                   |
| 272 | ROUND          | 1.3 | 11.03.08 | corrected an error with negative numbers                                                                                     |
| 272 | SIGMOID        | 1.1 | 11.03.08 | extended range of valid inputs to +inv / -inv                                                                                |
| 272 | FIB            | 1.2 | 10.03.08 | changed output to dint instead of real                                                                                       |
| 272 | FACT           | 1.3 | 10.03.08 | changed output of fact to dint to allow bigger values                                                                        |
| 272 | COTH           | 1.3 | 10.03.08 | extended range of valid inputs to +/- INV<br>changed formula for better accuracy                                             |
| 272 | TANH           | 1.2 | 10.03.08 | corrected an error in formula<br>extended range of valid inputs to +/- INV                                                   |
| 272 | RDM            | 1.5 | 10.03.08 | make sure last will be between 0 and 1 to avoid invalid results                                                              |
| 272 | _BUFFER_INSERT | 1.0 | 09.03.08 | New Module                                                                                                                   |

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|-----|-------------------|-----|----------|-----------------------------------------------------------------------------|
| 272 | CODE              | 1.0 | 09.03.08 | New Module                                                                  |
| 272 | HEAT_METER        | 1.2 | 06.03.08 | corrected an error in pulse mode                                            |
| 272 | ISC_UPPER         | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | ISC_LOWER         | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | ISC_HEX           | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | ISC_CTRL          | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | ISC_NUM           | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | ISC_ALPHA         | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | LOWERCASE         | 1.2 | 06.03.08 | added support for extended Ascii                                            |
| 272 | UPPERCASE         | 1.2 | 06.03.08 | added support for extended Ascii                                            |
| 272 | TO_LOWER          | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | TO_UPPER          | 1.0 | 06.03.08 | New Module                                                                  |
| 272 | GLOBAL CONSTANTS  |     | 06.03.08 | Added EXTENDED_ASCII                                                        |
| 272 | BUFFER_TO_STRING  | 1.0 | 05.03.08 | New Module                                                                  |
| 272 | BUFFER_SEARCH     | 1.0 | 05.03.08 | New Module                                                                  |
| 272 | _STRING_TO_BUFFER | 1.0 | 05.03.08 | New Module                                                                  |
| 272 | _BUFFER_INIT      | 1.0 | 05.03.08 | New Module                                                                  |
| 272 | _BUFFER_CLEAR     | 1.0 | 05.03.08 | New Module                                                                  |
| 272 | CAPITALIZE        | 1.0 | 04.03.08 | New Module                                                                  |
| 272 | IP4_CHECK         | 1.0 | 04.03.08 | New Module                                                                  |
| 272 | REAL_TO_STRF      | 1.3 | 04.03.08 | result is now rounded instead of trunc<br>corrected an error with reals < 1 |
| 272 | _ARRAY_SHUFFLE    | 1.0 | 04.03.08 | New Module                                                                  |
| 272 | HEX_TO_DWORD      | 1.2 | 04.03.08 | added support for a..f and return 0 for invalid string                      |
| 272 | EXEC              | 1.2 | 04.03.08 | add a 0 to the string if a '.' is at the end of the string                  |
| 272 | REPLACE_UML       | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_UPPER          | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_NUM            | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_LOWER          | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_IP4            | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_HEX            | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_CTRL           | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_ALPHA          | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IS_ALNUM          | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | IP4_DECODE        | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | COUNT_CHAR        | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | FINDB_NUM         | 1.1 | 29.02.08 | Improved code for higher speed                                              |
| 272 | FINDB_NONUM       | 1.1 | 29.02.08 | Improved code for higher speed                                              |
| 272 | FIND_CHAR         | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | FIND_CTRL         | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | FINDP             | 1.1 | 29.02.08 | ADDED MAX(pos,1) in loop initialization                                     |
| 272 | BIT_LOAD_B        | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | BIT_LOAD_DW       | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | BIT_LOAD_W        | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | RDM2              | 1.0 | 29.02.08 | New Module                                                                  |
| 272 | FINDB             | 1.2 | 29.02.08 | added findb := 0 for compatibility reasons                                  |
| 272 | FIND_NUM          | 1.1 | 29.02.08 | added input pos to start search at position                                 |
| 272 | FIND_NONUM        | 1.1 | 29.02.08 | added input pos to start search at position                                 |
| 271 | AOUT              | 1.4 | 23.02.08 | Improved code for higher speed                                              |
| 271 | AOUT1             | 1.0 | 23.02.08 | New Module                                                                  |
| 271 | AIN1              | 1.0 | 23.02.08 | New Module                                                                  |
| 271 | POLYNOM_INT       | 1.2 | 22.02.08 | Improved code for higher speed                                              |
| 271 | TRIANGLE_A        | 1.1 | 22.02.08 | Improved code for higher speed                                              |
| 271 | SPHERE_V          | 1.2 | 22.02.08 | Improved code for higher speed                                              |
| 271 | CONE_V            | 1.2 | 22.02.08 | Improved code for higher speed                                              |
| 271 | CIRCLE_C          | 1.1 | 22.02.08 | Improved code for higher speed                                              |
| 271 | CIRCLE_A          | 1.2 | 22.02.08 | Improved code for higher speed                                              |
| 271 | CABS              | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CACOS             | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CACOSH            | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CADD              | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CARG              | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CASIN             | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CASINH            | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CATAN             | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CATANH            | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CCON              | 1.0 | 21.02.08 | New Module                                                                  |
| 271 | CCOS              | 1.0 | 21.02.08 | New Module                                                                  |

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|-----|----------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 271 | CCOSH          | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CDIV           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CEXP           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CINV           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CLOG           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CMUL           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CPOL           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CPOW           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CSET           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CSIN           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CSINH          | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CSQRT          | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CSUB           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CTAN           | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | CTANH          | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | HYPOT          | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | TEMPERATURE    | 1.0 | 21.02.08 | New Module                                                                                                                                  |
| 271 | RTC_2          | 1.1 | 20.02.08 | Added millisecond set input                                                                                                                 |
| 271 | RTC_MS         | 1.1 | 20.02.08 | Added millisecond set input                                                                                                                 |
| 271 | WORD_OF_BYTE   | 1.2 | 19.02.08 | replaced + with or for better compatibility                                                                                                 |
| 271 | D_TRIG         | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | STACK_32       | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | STACK_16       | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | FIFO_32        | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | FIFO_16        | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | PARITY         | 1.3 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | CHECK_PARITY   | 1.3 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | BYTE_OF_BIT    | 1.1 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | REVERSE        | 1.1 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | BLIND_SECURITY | 1.2 | 12.02.08 | added config variables rain_up and wind_up<br>position for rain and wind is now configurable                                                |
| 27  | FT_INT         | 1.7 | 08.02.08 | Added interim variable to allow for changes smaller than 8 digits of output                                                                 |
| 27  | METER          | 1.1 | 08.02.08 | Use new version of ft_int and deleted unnecessary limits for integrators                                                                    |
| 27  | SWAP_BYTE2     | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | SWAP_BYTE      | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | REPLACE_ALL    | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | CHR            | 1.1 | 04.02.08 | return string would not be terminated properly                                                                                              |
| 27  | FINDP          | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | MIRROR         | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | TRIM1          | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | LOWERCASE      | 1.1 | 04.02.08 | improved performance<br>added support for Å, Ö, Ü                                                                                           |
| 27  | UPPERCASE      | 1.1 | 04.02.08 | improved performance<br>added support for Å, Ö, Ü                                                                                           |
| 27  | EASTER         | 1.2 | 03.02.08 | Modified code for better performance                                                                                                        |
| 27  | RDM            | 1.4 | 02.02.08 | changed algorithm to avoid non iec functions and guarantee more randomness                                                                  |
| 27  | BLIND_SCENE    | 1.1 | 29.01.08 | Replaced shr/shl with and for better performance                                                                                            |
| 27  | FT_RMP         | 1.4 | 25.01.08 | Improved algorithm for higher speed<br>allow KR and KF to be 0                                                                              |
| 27  | DCF77          | 1.7 | 23.01.08 | added sdt input which is used to initialize rtc and rtc1 during first cycle.                                                                |
| 27  | RTC_2          | 1.0 | 21.01.08 | New module                                                                                                                                  |
| 27  | RTC_MS         | 1.0 | 20.01.08 | New Module                                                                                                                                  |
| 27  | SUN_POS        | 1.2 | 18.01.08 | further performance improvements<br>only calculate once every 10 seconds                                                                    |
| 27  | SUN_TIME       |     | 17.01.08 | calculation is now only performed once a day                                                                                                |
| 27  | TIMER_1        | 1.6 | 17.01.08 | New code with better performance<br>Output is now only active during the time of day it is supposed to<br>no pulse is generated at power up |
| 27  | COUNT_BR       | 1.0 | 16.01.08 | New Module                                                                                                                                  |
| 27  | SELECT_8       | 1.0 | 16.01.08 | New Module                                                                                                                                  |
| 27  | INC            | 1.1 | 15.01.08 | Allow for negative increment                                                                                                                |
| 27  | RANGE_TO_WORD  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | RANGE_TO_BYTE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | WORD_TO_RANGE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | BYTE_TO_RANGE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | DEW_TEMP       | 1.4 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | LEAP_OF_DATE   | 1.2 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | DAY_OF_MONTH   | 1.4 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |

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| 27 | COTH           | 1.2 | 08.01.08 | Improved algorithm for higher speed                                     |
| 27 | ROUND          | 1.2 | 08.01.08 | Improved algorithm for higher speed                                     |
| 27 | SH_2           | 1.4 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | STAIR          | 1.2 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | SCALE_B8       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | SCALE_B4       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | SCALE_B2       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | GEN_SQR        | 1.3 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | GEN_SIN        | 1.3 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | TEMP_SI        | 1.4 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | TEMP_NI        | 1.4 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | BAR_GRAPH      | 1.2 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | KMH_TO_MS      | 1.1 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | F_TO_C         | 1.1 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | SUN_POS        | 1.1 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | SUN_TIME       | 1.3 | 06.01.08 | Improved algorithm for higher speed                                     |
| 27 | _ARRAY_SORT    | 1.5 | 05.01.08 | Improved algorithm for higher speed                                     |
| 27 | SINH           | 1.2 | 05.01.08 | Improved algorithm for higher speed                                     |
| 27 | COSH           | 1.2 | 05.01.08 | Improved algorithm for higher speed                                     |
| 27 | ATANH          | 1.1 | 05.01.08 | Improved algorithm for higher speed                                     |
| 27 | ACOTH          | 1.1 | 05.01.08 | Improved algorithm for higher speed                                     |
| 27 | HYST_3         | 1.2 | 05.01.08 | Improved code for higher performance                                    |
| 27 | HYST_2         | 1.1 | 05.01.08 | Improved code for higher performance                                    |
| 27 | FT_PT1         | 1.7 | 05.01.08 | Improved code for higher performance                                    |
| 27 | FT_PID         | 1.6 | 05.01.08 | Improved code for higher performance                                    |
| 27 | FT_PI          | 1.2 | 05.01.08 | Improved performance                                                    |
| 27 | FT_INT         | 1.6 | 05.01.08 | Improved performance                                                    |
| 27 | TEMP_PT        | 1.5 | 05.01.08 | replaced /2 with * 0.5 for better performance                           |
| 27 | RDM            | 1.3 | 05.01.08 | Corrected an error with high values of sps timer                        |
| 27 | PERIOD         | 1.2 | 04.01.08 | Improved performance                                                    |
| 27 | DAY_OF_YEAR    | 1.2 | 04.01.08 | Improved performance                                                    |
| 27 | YEAR_BEGIN     | 1.1 | 04.01.08 | formula for step7 was incorrect during leap years                       |
| 27 | SET_DATE       | 1.4 | 03.01.08 | Improved performance                                                    |
| 27 | IRTRANS_DECODE | 1.0 | 02.01.08 | New Module                                                              |
| 27 | IRTRANS_8      | 1.0 | 02.01.08 | New Module                                                              |
| 27 | IRTRANS_4      | 1.0 | 02.01.08 | New Module                                                              |
| 27 | IRTRANS_1      | 1.0 | 02.01.08 | New Module                                                              |
| 27 | DRIVER_4       | 1.0 | 02.01.08 | New Module                                                              |
| 27 | DRIVER_1       | 1.0 | 02.01.08 | New Module                                                              |
| 27 | CLK_DIV        | 1.1 | 02.01.08 | Improved performance                                                    |
| 27 | BYTE_OF_DWORD  | 1.1 | 02.01.08 | Improved performance                                                    |
| 27 | WORD_OF_DWORD  | 1.1 | 02.01.08 | Improved performance                                                    |
| 27 | FIB            | 1.1 | 02.01.08 | deleted unused variable pt                                              |
| 27 | WORD_OF_BYTE   | 1.1 | 02.01.08 | Improved performance                                                    |
| 27 | CRC_GEN        | 1.3 | 02.01.08 | Improved performance                                                    |
| 27 | DWORD_OF_BYTE  | 1.1 | 02.01.08 | Improved performance                                                    |
| 26 | BLIND_INPUT    | 1.2 | 30.12.07 | added click_timeout variable and function                               |
| 26 | FIB            | 1.0 | 27.12.07 | New Module                                                              |
| 26 | WATER_CP       | 1.1 | 27.12.07 | Increased accuracy                                                      |
| 26 | SHR1           | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | SHL1           | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | LTCH_4         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | LTCH           | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | FF_JKE         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | FF_DRE         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | FF_D4E         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | FF_D2E         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | STAIR          | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | STAIR2         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | HYST_3         | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | ACTUATOR_2P    | 1.5 | 27.12.07 | changed code for better performance                                     |
| 26 | LINEAR_INT     | 1.1 | 27.12.07 | changed code for better performance                                     |
| 26 | STORE_8        | 1.0 | 26.12.07 | New Module                                                              |
| 26 | MESSAGE_8      | 1.0 | 26.12.07 | New Module                                                              |
| 26 | FADE           | 1.0 | 26.12.07 | New Module                                                              |
| 26 | RMP_B          | 1.6 | 25.12.07 | changed code for better performance                                     |
| 26 | RMP_W          | 1.7 | 25.12.07 | corrected an error while step response was too slow for fast rist times |

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|     |               |     |          |                                                                              |
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| 26  | HEAT_METER    | 1.1 | 25.12.07 | added config variable return_meter<br>added output C for current consumption |
| 26  | MATRIX        | 1.2 | 23.12.07 | added exit statement in for loop instead of i:=5                             |
| 26  | WEEKDAY       | 1.2 | 23.12.07 | correction for step7                                                         |
| 26  | TEMP_PT       | 1.4 | 23.12.07 | avoid a negative square root if input values are wrong                       |
| 26  | YEAR          | 1.2 | 23.12.07 | changed code for better performance                                          |
| 26  | WORK_WEEK     | 1.2 | 19.12.07 | changed code for better performance<br>changed code to comply with ISO8601   |
| 26  | YEAR_BEGIN    | 1.0 | 19.12.07 | New Module                                                                   |
| 26  | TEMP_NI       | 1.3 | 17.12.07 | changed code for better performance                                          |
| 26  | POLYNOM_INT   | 1.1 | 17.12.07 | init makes no sense for a function                                           |
| 26  | TEMP_SI       | 1.3 | 17.12.07 | changed code for better performance                                          |
| 26  | YEAR          | 1.2 | 16.12.07 | changed code for better performance                                          |
| 26  | SET_DATE      | 1.3 | 16.12.07 | changed code for better performance                                          |
| 26  | OSCAT_VERSION | 1.1 | 16.12.07 | added possibility to return date and version depending on in                 |
| 26  | CLK_8MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                                     |
| 26  | CLK_4MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                                     |
| 26  | CLK_2MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                                     |
| 26  | CLK_1MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                                     |
| 26  | CLK_N         | 1.0 | 16.12.07 | New Module                                                                   |
| 26  | EASTER        | 1.1 | 15.12.07 | changed code for better performance                                          |
| 26  | REAL_TO_STRF  | 1.2 | 15.12.07 | changed code for better performance                                          |
| 26  | DWORD_TO_STRH | 1.2 | 15.12.07 | changed code for better performance                                          |
| 26  | DWORD_TO_STRB | 1.1 | 15.12.07 | changed code for better performance                                          |
| 26  | BYTE_TO_STRH  | 1.2 | 15.12.07 | changed code for better performance                                          |
| 26  | BYTE_TO_STRB  | 1.1 | 15.12.07 | changed code for better performance                                          |
| 26  | FINDB         | 1.1 | 15.12.07 | changed code for better performance                                          |
| 26  | TICKER        | 1.1 | 15.12.07 | step now starts at 1 instaed of 0                                            |
| 26  | _ARRAY_SORT   | 1.4 | 15.12.07 | added function return = true                                                 |
| 26  | SIGNAL_4      | 1.0 | 13.12.07 | New Module                                                                   |
| 26  | SIGNAL        | 1.0 | 13.12.07 | New Module                                                                   |
| 26  | INT_TO_BCD    | 1.0 | 13.12.07 | New Module                                                                   |
| 26  | BCD_TO_INT    | 1.0 | 13.12.07 | New Module                                                                   |
| 26  | ARRAY_VAR     | 1.1 | 12.12.07 | changed code for better performance                                          |
| 26  | ARRAY_AVG     | 1.1 | 12.12.07 | changed code for better performance                                          |
| 26  | ARRAY_VAR     | 1.1 | 12.12.07 | changed code for better performance                                          |
| 26  | ARRAY_TREND   | 1.1 | 12.12.07 | changed code for better performance                                          |
| 26  | V3_ZANG       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_YANG       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_XANG       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_XPRO       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_SMUL       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_REV        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_PAR        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_NORM       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_DPRO       | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_SUB        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_ADD        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_ANG        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | V3_ABS        | 1.0 | 11.12.07 | New Module                                                                   |
| 26  | LANGEVIN      | 1.0 | 10.12.07 | New Module                                                                   |
| 26  | SIGMOID       | 1.0 | 10.12.07 | New Module                                                                   |
| 26  | GAMMA         | 1.0 | 10.12.07 | New Module                                                                   |
| 26  | FACT          | 1.2 | 10.12.07 | start value for i has changed to 2 for better performance                    |
| 26  | TONOF         | 1.0 | 10.12.07 | New Module                                                                   |
| 26  | PWM_PW        | 1.3 | 09.12.07 | Corrected a problem with F = 0                                               |
| 252 | DCF77         | 1.6 | 08.12.07 | corrected a problem with time_zone_offset                                    |
| 251 | LTIME_TO_UTC  | 1.3 | 08.12.07 | corrected a problem with time_zone_offset                                    |
| 251 | LTIME         | 1.6 | 08.12.07 | corrected a problem with time_zone_offset                                    |
| 251 | LOCAL_TIME    | 1.4 | 08.12.07 | corrected a problem with time_zone_offset                                    |
| 25  | ESR_COLLECT   | 1.1 | 08.12.07 | Added reset input                                                            |
| 25  | ESR_MON_R4    | 1.2 | 08.12.07 | corrected an error while esr type would not be set                           |
| 25  | CHECK_PARITY  | 1.2 | 08.12.07 | changed code for better performance                                          |
| 25  | PARITY        | 1.2 | 08.12.07 | changed code for better performance                                          |
| 25  | WCT           | 1.1 | 07.12.07 | changed code for better performance                                          |
| 25  | SPHERE_V      | 1.1 | 04.12.07 | changed code for better performance                                          |
| 25  | CONE_V        | 1.1 | 04.12.07 | changed code for better performance                                          |
| 25  | CIRCLE_A      | 1.1 | 04.12.07 | changed code for better performance                                          |

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|    |               |     |          |                                                                                                                       |
|----|---------------|-----|----------|-----------------------------------------------------------------------------------------------------------------------|
| 25 | EXP_N         | 1.0 | 04.12.07 | New Module                                                                                                            |
| 25 | TICKER        | 1.0 | 04.12.07 | New Module                                                                                                            |
| 25 | AIR_DENSITY   | 1.1 | 04.12.07 | changed code for better performance                                                                                   |
| 25 | SQRTN         | 1.1 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | ROUND         | 1.1 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | ASINH         | 1.1 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | ACOSH         | 1.1 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | AOUT          | 1.3 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | SDD           | 1.1 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | DEW_TEMP      | 1.3 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | DEW_CON       | 1.2 | 02.12.07 | changed code for better performance<br>output is now in Gramms / Qubic meter                                          |
| 25 | AIR_ENTHALPY  | 1.2 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | EXP10         | 1.0 | 02.12.07 | New Module                                                                                                            |
| 25 | AIN           | 1.4 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RMP_W         | 1.6 | 02.12.07 | corrected an error in calculation of step response                                                                    |
| 25 | TEMP_PT       | 1.3 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | SENSOR_INT    | 1.1 | 02.12.07 | Corrected an error in algorithm                                                                                       |
| 25 | RES_SI        | 1.1 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RES_PT        | 1.2 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RES_NI        | 1.2 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | ONTIME        | 1.4 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | BAR_GRAPH     | 1.1 | 02.12.07 | changed code to improove performance                                                                                  |
| 25 | FT_INT        | 1.5 | 02.12.07 | changed code to improove performance                                                                                  |
| 25 | WATER_DENSITY | 1.1 | 02.12.07 | new code for better performance                                                                                       |
| 25 | HEAT_INDEX    | 1.2 | 02.12.07 | new code for better performance                                                                                       |
| 25 | DEW_RH        | 1.1 | 02.12.07 | new code for better performance                                                                                       |
| 25 | DST           | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | COTH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | COSH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | TANH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SINH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SIGN_R        | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SIGN_I        | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | EVEN          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | FT_PT2        | 1.3 | 30.11.07 | changed out to be K * in during initialization<br>avoid divide by 0 if TN = 0                                         |
| 25 | FT_PT1        | 1.6 | 30.11.07 | changed out to be K * in during initialization                                                                        |
| 25 | PWM_PW        | 1.2 | 27.11.07 | avoid divide by 0 when F = 0                                                                                          |
| 25 | PWM_DC        | 1.2 | 27.11.07 | avoid divide by 0 when F = 0                                                                                          |
| 25 | GEN_SIN       | 1.2 | 27.11.07 | avoid divide by 0 when pt = 0                                                                                         |
| 25 | GEN_RMP       | 1.2 | 27.11.07 | avoid divide by 0 when pt = 0                                                                                         |
| 24 | BLIND_SHADE   | 1.2 | 24.11.07 | added shade_delay to avoid constant up and down if sun is out parcially                                               |
| 24 | HOLIDAY       | 1.3 | 24.11.07 | changes F_use of Buß_und_Bettag to 0 because this is no official holiday                                              |
| 24 | REAL_TO_STRF  | 1.1 | 20.11.07 | when N=0 there will be no dot at the end of the string                                                                |
| 24 | GEN_RDT       | 1.0 | 20.11.07 | New Module                                                                                                            |
| 24 | RDM           | 1.2 | 20.11.07 | changed code of temp calculation to avoid overflow in modr due to resucit would not fit<br>DINT for high timer values |
| 24 | ACTUATOR_2P   | 1.4 | 19.11.07 | replaced left over statement time() with tx                                                                           |
| 24 | MIX           | 1.0 | 19.11.07 | New Module                                                                                                            |
| 24 | HEX_TO_DWORD  | 1.1 | 19.11.07 | changed type of function from int to dword                                                                            |
| 24 | HEAT_METER    | 1.0 | 17.11.07 | New Module                                                                                                            |
| 24 | _ARRAY_INIT   | 1.2 | 14.11.07 | changed size calculation to be more efficient                                                                         |
| 24 | _ARRAY_SORT   | 1.3 | 14.11.07 | changed size calculation to be more efficient                                                                         |
| 24 | _ARRAY_MEDIAN | 1.4 | 14.11.07 | corrected a problem with size calculation                                                                             |
| 24 | T_PLC_MS      | 1.1 | 12.11.07 | added temporaray variable tx because some compilers could not handle time() as an<br>argument                         |
| 24 | T_PLC_US      | 1.1 | 12.11.07 | added temporaray variable tx because some compilers could not handle time() as an<br>argument                         |
| 24 | LTIME_TO_UTC  | 1.2 | 12.11.07 | changed Type of time_zone_offset from time to int<br>to allow for time zones with negative offset                     |
| 24 | LTIME         | 1.5 | 12.11.07 | changed Type of time_zone_offset from time to int<br>to allow for time zones with negative offset                     |
| 24 | LOCAL_TIME    | 1.3 | 12.11.07 | changed time_zone_offset from time to integer to allow for negative offset time zones                                 |
| 24 | DCF77         | 1.5 | 12.11.07 | changed time_offset from time to integer to allow for negative offset time zones                                      |
| 24 | RDM           | 1.1 | 11.11.07 | changed time() into t_plc_ms()                                                                                        |
| 24 | MUL_ADD       | 1.1 | 11.11.07 | deleted preset values for K and O this makes no sense for a function                                                  |
| 24 | NEG           | 1.0 | 11.11.07 | New Module                                                                                                            |

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|    |                  |     |          |                                                                                                                                                   |
|----|------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | SGN              | 1.1 | 11.11.07 | changed type of function from real to int                                                                                                         |
| 24 | LTIME_TO_UTC     | 1.1 | 05.11.07 | replaced literal constant with variable because of error in möller ecp4 compiler                                                                  |
| 24 | CONTROL_SET2     | 1.0 | 04.11.07 | New Module                                                                                                                                        |
| 24 | CONTROL_SET1     | 1.0 | 04.11.07 | New Module                                                                                                                                        |
| 24 | TUNE             | 1.0 | 04.11.07 | New Module                                                                                                                                        |
| 24 | TUNE2            | 1.0 | 03.11.07 | New Module                                                                                                                                        |
| 24 | PARSET2          | 1.0 | 03.11.07 | New Module                                                                                                                                        |
| 24 | FT_PI            | 1.1 | 03.11.07 | added noise input to filter noise<br>output diff added<br>set limit output false when output is within limits<br>overflow was not set correctly   |
| 24 | FT_PID           | 1.5 | 03.11.07 | added noise input to filter noise,<br>output diff added,<br>set limit output false when output is within limits<br>overflow was not set correctly |
| 24 | SET_TOD          | 1.3 | 02.11.07 | changed dword to DINT in calculation to avoid warnings with some compilers                                                                        |
| 24 | PARSET           | 1.0 | 02.11.07 | New Module                                                                                                                                        |
| 24 | DEAD_BAND        | 1.0 | 02.11.07 | New Module                                                                                                                                        |
| 24 | AIR_ENTHALPY     | 1.1 | 02.11.07 | deleted unused variable RF                                                                                                                        |
| 23 | DATE_ADD         | 1.2 | 01.11.07 | added int_to_dword statements to avoid possible overrun with möller ecp4                                                                          |
| 23 | SET_TOD          | 1.2 | 01.11.07 | changed coding to avoid possible overrun situation on möller ecp4                                                                                 |
| 23 | SET_DT           | 1.4 | 01.11.07 | converted hour type integer to dword in calculation to avoid overrun on möller ecp4                                                               |
| 23 | WINDOW2          | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | FT_PID           | 1.4 | 31.10.07 | Rewritten to avoid error when one of the limits is 0                                                                                              |
| 23 | FT_PI            | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | AIR_ENTHALPY     | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | HOLIDAY          | 1.2 | 31.10.07 | changed holiday definition from constant to input constant                                                                                        |
| 23 | BLIND_CONTROL    | 1.3 | 30.10.07 | added lockout configuration time for motor direction change                                                                                       |
| 23 | AIR_DENSITY      | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | SDD              | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | WATER_ENTHALPY   | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | FT_DERIV         | 1.3 | 29.10.07 | prohibit calculation when tx - last = 0 to avoid div 0 and increase accuracy on fast systems                                                      |
| 23 | FT_INT           | 1.4 | 29.10.07 | changed code so int will not be called is time difference is 0 to increase accuracy on systems with cycle times below 1ms                         |
| 23 | BLIND_SCENE      | 1.0 | 24.10.07 | new module                                                                                                                                        |
| 23 | LTIME            | 1.4 | 24.10.07 | changed code because the execution every 100ms can cause major problems<br>if the supplied time was not correct at start. use of new dst function |
| 23 | LOCAL_TIME       | 1.2 | 24.10.07 | changed dst calculation because function dst was upgraded no error in local_time only a change in DST                                             |
| 23 | DCF77            | 1.4 | 24.10.07 | changed dst calculation because function dst was upgraded no error in DCF77 only a change in DST                                                  |
| 23 | DST              | 1.1 | 24.10.07 | deleted time_zone_offset input because dst is generally at 01:00 utc and not mesz<br>uk starts 01:00 utc and also greece                          |
| 23 | FT_PT1           | 1.5 | 23.10.07 | added out := in to the init statements                                                                                                            |
| 23 | BLIND_SHADE      | 1.1 | 19.10.07 | manual mode was not detected because outputs were checked and not inputs UP and DN                                                                |
| 23 | CONE_V           | 1.0 | 17.10.07 | New Module                                                                                                                                        |
| 22 | GLOBAL CONSTANTS |     | 16.10.07 | Renamed PI in PI1 for better compatibility to other libraries                                                                                     |
| 22 | DEG              | 1.1 | 16.10.07 | Deg will become 0 when it reaches 360                                                                                                             |
| 22 | RAD              | 1.1 | 16.10.07 | Rad will become 0 when it reaches 2PI                                                                                                             |
| 22 | TRIANGLE_A       | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | SPHERE_V         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | ELLIPSE_C        | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | ELLIPSE_A        | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | CIRCLE_C         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | CIRCLE_A         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | SGN              | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | BLIND_CONTROL    | 1.2 | 16.10.07 | Ignore angle when T_ANGLE < T#100ms                                                                                                               |
| 22 | BLIND_SHADE      | 1.0 | 15.10.07 | New Module                                                                                                                                        |
| 22 | CRC_GEN          | 1.2 | 09.10.07 | Added init values for start value and final XOR<br>added reflect_in and reflect_out feature                                                       |
| 22 | REVERSE          | 1.0 | 09.10.07 | New Module                                                                                                                                        |
| 22 | SET_DT           | 1.3 | 08.10.07 | deleted unused variables count and leap                                                                                                           |
| 22 | DAY_OF_MONTH     | 1.3 | 08.10.07 | deleted unused variable yr                                                                                                                        |
| 22 | _ARRAY_MEDIAN    | 1.3 | 08.10.07 | deleted unused variables m and temp                                                                                                               |
| 22 | POLYNOM_INT      | 1.0 | 08.10.07 | New Module                                                                                                                                        |
| 22 | LINEAR_INT       | 1.0 | 07.10.07 | New Module                                                                                                                                        |

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|    |                  |     |          |                                                                                                                                                                                                   |
|----|------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | BLIND_SECURITY   | 1.1 | 06.10.07 | Added pos and angle inputs and outputs<br>actions are now configurable                                                                                                                            |
| 22 | BLIND_INPUT      | 1.1 | 06.10.07 | Renamed output ps and as to po and ao<br>added click mode                                                                                                                                         |
| 22 | BLIND_NIGHT      | 1.1 | 06.10.07 | added pos and angle inputs and outputs<br>night position and angle can now be configured<br>any manual operation at night will cancel night operation                                             |
| 22 | BLIND_CONTROL    | 1.1 | 06.10.07 | Integrated blind_actuator into BLIND_CONTROL                                                                                                                                                      |
| 22 | BLIND_ACTUATOR   | 1.1 | 06.10.07 | Rearranged outputs                                                                                                                                                                                |
| 22 | RMP_W            | 1.5 | 05.10.07 | Corrected a problem with pt = 0ms                                                                                                                                                                 |
| 22 | RMP_B            | 1.5 | 05.10.07 | Corrected a problem with pt = 0ms                                                                                                                                                                 |
| 22 | HEX_TO_DWORD     | 1.0 | 02.10.07 | New Module                                                                                                                                                                                        |
| 21 | DAY_OF_DATE      | 1.1 | 01.10.07 | Added step7 compatibility                                                                                                                                                                         |
| 21 | MONTH            | 1.1 | 01.10.07 | New coding for much better performance, no string functions used                                                                                                                                  |
| 21 | SET_DT           | 1.2 | 01.10.07 | Added step7 compatibility                                                                                                                                                                         |
| 21 | SET_DATE         | 1.2 | 01.10.07 | added step7 compatibility, now call function set_date                                                                                                                                             |
| 21 | TIMER_1          | 1.5 | 01.10.07 | Corrected a problem , timer would not generate a pulse over midnight                                                                                                                              |
| 21 | DAY_OF_MONTH     | 1.2 | 01.10.07 | Corrected an error in algorithm.                                                                                                                                                                  |
| 21 | LEAP_YEAR        | 1.2 | 01.10.07 | Improved code for higher performance                                                                                                                                                              |
| 21 | DAY_OF_YEAR      | 1.1 | 01.10.07 | Added compatibility to step7                                                                                                                                                                      |
| 21 | YEAR             | 1.1 | 01.10.07 | Corrected error and added compatibility to Step7                                                                                                                                                  |
| 21 | GLOBAL CONSTANTS |     | 01.10.07 | Added Setup Parameter STEP7                                                                                                                                                                       |
| 21 | BLIND_NIGHT      | 1.0 | 30.09.07 | New Module                                                                                                                                                                                        |
| 21 | BLIND_SHADER     | 1.0 | 30.09.07 | New Module                                                                                                                                                                                        |
| 21 | BLIND_INPUT      | 1.0 | 29.09.07 | New Module                                                                                                                                                                                        |
| 21 | BLIND_CONTROL    | 1.0 | 29.09.07 | New Module                                                                                                                                                                                        |
| 21 | BLIND_SECURITY   | 1.0 | 29.09.07 | New Module                                                                                                                                                                                        |
| 21 | BLIND_ACTUATOR   | 1.0 | 29.09.07 | New Module                                                                                                                                                                                        |
| 21 | INTERLOCK        | 1.0 | 28.09.07 | New Module                                                                                                                                                                                        |
| 21 | RMP_W            | 1.4 | 28.09.07 | Added output high and low, coorrected error while Busy was displayed when en was<br>false<br>improoved performance and timing accuracy<br>added init statements to avoid wrong startup conditions |
| 21 | RMP_B            | 1.4 | 28.09.07 | Added output high and low, coorrected error while Busy was displayed when en was<br>false<br>improoved performance and timing accuracy<br>added init statements to avoid wrong startup conditions |
| 21 | _ARRAY_MEDIAN    | 1.2 | 22.09.07 | chagned code to use array_sort<br>Improved performance                                                                                                                                            |
| 21 | _ARRAY_SORT      | 1.2 | 22.09.07 | new algorithm for better performance                                                                                                                                                              |
| 21 | PERIOD           | 1.1 | 20.09.07 | corrected a problem with leap year                                                                                                                                                                |
| 20 | SET_DT           | 1.1 | 19.09.07 | Corrected leap year calculation                                                                                                                                                                   |
| 20 | SET_DATE         | 1.1 | 19.09.07 | Corrected leap year calculation                                                                                                                                                                   |
| 20 | PERIOD           | 1.0 | 19.09.07 | New Module                                                                                                                                                                                        |
| 20 | DCF77            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | ESR_MON_X8       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | ESR_MON_R4       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | ESR_MON_B8       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | FT_RMP           | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | TP_R             | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | SEQUENCE_8       | 1.4 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | SEQUENCE_4       | 1.4 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | GEN_SQ           | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_PULSE        | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_PRG          | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_8MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_4MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_2MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | CLK_1MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | SH_2             | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | SH_1             | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | RMP_W            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | RMP_SOFT         | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | RMP_B            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | GEN_SQ           | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | GEN_SIN          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | GEN_RMP          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                           |
| 20 | DAYS_DELTA       | 1.1 | 16.09.07 | Corrected an error in algorithm                                                                                                                                                                   |
| 20 | DAY_OF_DATE      | 1.0 | 16.09.07 | New Module                                                                                                                                                                                        |

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|    |               |     |          |                                                                                                         |
|----|---------------|-----|----------|---------------------------------------------------------------------------------------------------------|
| 20 | METER_STAT    | 1.0 | 16.09.07 | New Module                                                                                              |
| 20 | METER         | 1.0 | 16.09.07 | New Module                                                                                              |
| 20 | GEN_RDM       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ONTIME        | 1.3 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_TX          | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_T           | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_D           | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_TN64       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_TN16       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_PT2        | 1.2 | 16.09.07 | Deleted unused code section for system time reading and unnecessary startup code                        |
| 20 | FT_PT1        | 1.4 | 16.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | CYCLE_TIME    | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_INT        | 1.3 | 15.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | FT_DERIV      | 1.2 | 15.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | FT_PROFILE    | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TEMP_EXT      | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TIMER_2       | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TIMER_1       | 1.4 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | SWITCH_I      | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | PULSE_LENGTH  | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | DIMM_I        | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | CLICK         | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_UD   | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_PUMP | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_3P   | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_2P   | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | SHL1          | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | SHR1          | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | TOGGLE        | 1.0 | 13.09.07 | New Module                                                                                              |
| 20 | T_PLC_US      | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | T_PLC_MS      | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | AOUT          | 1.2 | 13.09.07 | changed code to avoid warning under codesys 2.3.8.1                                                     |
| 20 | AIN           | 1.3 | 13.09.07 | changed code to avoid warning under codesys 2.3.8.1                                                     |
| 20 | RES_NI        | 1.1 | 13.09.07 | Changed coding for better performance                                                                   |
| 20 | DEW_TEMP      | 1.2 | 13.09.07 | Changed algorithm to improve performance                                                                |
| 20 | DEW_CON       | 1.1 | 13.09.07 | Changed algorithm to improve performance                                                                |
| 20 | WATER_DENSITY | 1.0 | 13.09.07 | New Module                                                                                              |
| 20 | DWORD_TO_STRH | 1.1 | 11.09.07 | changed coding for compatibility with twincat, under twincat concat cannot have a function as argument. |
| 20 | BYTE_TO_STRH  | 1.1 | 11.09.07 | changed coding for compatibility with twincat, under twincat concat cannot have a function as argument. |
| 20 | SET_TOD       | 1.1 | 11.09.07 | changed coding to avoid a compiler warning under twincat.                                               |
| 20 | MATRIX        | 1.1 | 11.09.07 | deleted unused variables k and old_code                                                                 |
| 20 | CRC_GEN       | 1.1 | 11.09.07 | Deleted unused variable i                                                                               |
| 20 | CHECK_PARITY  | 1.1 | 10.09.07 | Changed algorithm to improve performance                                                                |
| 20 | PARITY        | 1.1 | 10.09.07 | Changed algorithm to improve performance                                                                |
| 20 | TEMP_SII      | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time and corrected an error in formula             |
| 20 | TEMP_PT       | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time                                               |
| 20 | TEMP_NI       | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time                                               |
| 20 | BIT_COUNT     | 1.1 | 10.09.07 | Improved performance                                                                                    |
| 20 | SH_2          | 1.2 | 10.09.07 | Corrected an error while index would be out of array if N = 0;                                          |
| 20 | D_TRIG        | 1.0 | 04.09.07 | New Module                                                                                              |
| 18 | ONTIME        | 1.2 | 02.08.07 | Corrected an error while ontime would stop counting time after a certain time is reached                |
| 17 | LTIME_TO_UTC  | 1.0 | 05.07.07 | New Module                                                                                              |
| 17 | BIT_COUNT     | 1.0 | 05.07.07 | New Module                                                                                              |
| 17 | MATRIX        | 1.0 | 10.06.07 | New Module                                                                                              |
| 17 | CRC_CHECK     | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | CRC_GEN       | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | BYTE_TO_STRB  | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | BYTE_TO_STRH  | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | DWORD_TO_STRB | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | DWORD_TO_STRH | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | CLICK         | 1.1 | 18.05.07 | Corrected an error while Click would stop working when t_reconfig expired                               |
| 16 | PRESSURE      | 1.0 | 27.03.07 | New Module                                                                                              |
| 16 | ARRAY_SDV     | 1.0 | 01.04.07 | New Module                                                                                              |
| 16 | LTIME         | 1.3 | 18.03.07 | changed code, dst would not work during first cycle.                                                    |

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| 16 | DEW_TEMP         | 1.1 | 29.04.07 | Changed code when RH = 0, not to execute log(0)                                                                                                                                            |
| 16 | GLOBAL CONSTANTS |     | 02.05.07 | Added UG, universal gas constant                                                                                                                                                           |
| 16 | ARRAY_VAR        | 1.0 | 01.04.07 | New Module                                                                                                                                                                                 |
| 16 | TIMER_1          | 1.3 | 14.04.07 | Corrected an error while output would not go active when start year = 1970<br>increased accuracy by checking every 100ms<br>added init section to compensate for timer overflow at startup |
| 16 | DATE_ADD         | 1.1 | 12.04.07 | corrected an error while date would be incorrect when year = 0                                                                                                                             |
| 16 | M_TX             | 1.1 | 02.05.07 | Corrected wrong measurements during startup<br>added output ET (elapsed time)<br>added Reset input<br>added tmax input                                                                     |
| 16 | LENGTH           | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | ASTRO            | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | ENERGY           | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | DEW_CON          | 1.0 | 29.04.07 | New Module                                                                                                                                                                                 |
| 16 | CYCLE_TIME       | 1.2 | 10.03.07 | changed initialization of ct_min to t#10h for compatibility with siemens s7                                                                                                                |
| 16 | SPEED            | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | M_D              | 1.0 | 02.05.07 | New Module                                                                                                                                                                                 |
| 16 | M_T              | 1.1 | 02.05.07 | Added init variable to avoid unreasonable result if falling edge occurs first<br>added ET (elapsed Time) output<br>added reset input                                                       |
| 16 | DEW_RH           | 1.0 | 29.04.07 | New Module                                                                                                                                                                                 |
| 16 | SUN_TIME         | 1.2 | 17.04.07 | corrected error while sun:midday would not be corrected for longitude.                                                                                                                     |
| 15 | MUL_ADD          | 1.0 | 08.02.07 | New Module                                                                                                                                                                                 |
| 15 | CLK_PRG          | 1.1 | 25.02.07 | rewritten moule in ST for better performance<br>PT can noe be changed during runtime                                                                                                       |
| 15 | DEC_8            | 1.1 | 03.03.07 | rewritten moule in ST for better performance                                                                                                                                               |
| 15 | FT_RMP           | 1.2 | 12.02.07 | Added INIT to prevent erroneous startup conditions                                                                                                                                         |
| 15 | GEN_RDM          | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | FT_PID           | 1.3 | 03.03.07 | Added default values for inputs KP, TN, TV, LIMIT_H and LIMIT_L                                                                                                                            |
| 15 | CEIL             | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | FT_PT1           | 1.3 | 27.02.07 | Output will be set to input during initial startup                                                                                                                                         |
| 15 | CLICK            | 1.0 | 15.02.07 | New Module „click“                                                                                                                                                                         |
| 15 | PULSE_LENGTH     | 1.0 | 15.02.07 | New Module                                                                                                                                                                                 |
| 15 | FACT             | 1.1 | 04.03.07 | Changed a critical erro while loop was indefinite                                                                                                                                          |
| 15 | GEN_RMP          | 1.0 | 03.03.07 | New Module                                                                                                                                                                                 |
| 15 | FLOOR            | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | DEAD_ZONE_2      | 1.0 | 12.02.07 | New Module                                                                                                                                                                                 |
| 15 | SIGN_I           | 1.1 | 03.03.07 | changed method of function for better compatibility to other systems                                                                                                                       |
| 15 | GLOBAL CONSTANTS |     | 03.03.07 | Increased accuracy of PI2<br>renamed T0 to TK for better compatibility with other Libraries<br>renamed C to C0 for better compatibility                                                    |
| 15 | WCT              | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | HOLIDAYEAT_INDEX | 1.1 | 27.02.07 | deleted unused variables rh3 and T3                                                                                                                                                        |
| 15 | PWM_DC           | 1.1 | 25.02.07 | rewritten moule in ST for better performance                                                                                                                                               |
| 15 | F_LAMP           | 1.1 | 08.02.07 | New Module                                                                                                                                                                                 |
| 15 | HOLIDAY          | 1.1 | 27.02.07 | deleted unused variable init                                                                                                                                                               |
| 15 | ONTIME           | 1.1 | 22.02.07 | Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system like WAGO's LON controllers                                                   |
| 15 | TIMER_2          | 1.1 | 22.02.07 | Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system like WAGO's LON controllers                                                   |
| 15 | _ARRAY_MEDIAN    | 1.1 | 03.03.07 | Corrected an error in median calculation                                                                                                                                                   |
| 15 | DEC_4            | 1.1 | 03.03.07 | rewritten moule in ST for better performance                                                                                                                                               |
| 15 | GEN_SQR          | 1.1 | 12.02.07 | Added default value for dc = 0.5                                                                                                                                                           |
| 15 | SWITCH_X         | 1.0 | 15.02.07 | New Module                                                                                                                                                                                 |
| 15 | CALIBRATE        | 1.2 | 22.02.07 | Chaged VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system                                                                                |
| 15 | DEC_2            | 1.1 | 03.03.07 | rewritten moule in ST for better performance                                                                                                                                               |
| 15 | FT_PROFILE       | 1.0 | 27.02.07 | New Module                                                                                                                                                                                 |
| 15 | INC              | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | PWM_PW           | 1.1 | 25.02.07 | rewritten moule in ST for better performance                                                                                                                                               |
| 15 | DEAD_ZONE        | 1.0 | 12.02.07 | New Module                                                                                                                                                                                 |
| 15 | BAR_GRAPH        | 1.0 | 22.02.07 | New Module                                                                                                                                                                                 |
| 15 | BOILER           | 1.0 | 27.02.07 | New Module                                                                                                                                                                                 |
| 15 | DCF_77           | 1.2 | 26.02.07 | Chaqnge statements to avoid a compiler error if wago 16 bit lon controller is used                                                                                                         |
| 14 | MUX_R2           | 1.0 | 19.01.07 | New Module                                                                                                                                                                                 |
| 14 | DIMM_I           | 1.1 | 22.01.07 | Deleted unused structure „ramp“                                                                                                                                                            |
| 14 | F_TO_OM          | 1.0 | 22.01.07 | New Module                                                                                                                                                                                 |
| 14 | ESR_MON_X8       | 1.0 | 26.01.07 | New Module                                                                                                                                                                                 |

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| 14 | ASINH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | DAY_OF_MONTH  | 1.1 | 22.01.07 | deleted unused variable day_in_year and day_in_year_begin                                                                                                                                                                           |
| 14 | Heat_Temp     | 1.0 | 27.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | DIMM_I        | 1.2 | 02.02.07 | Changed method to read time() only at the beginning of the module for better consistency when a higher priority task interrupts „dimm_i“<br>added code in case structure for state 1 to allow for short one cycle long input pulses |
| 14 | SEQUENCE_4    | 1.3 | 17.01.07 | Changed output fault to ESR compatible output STATUS<br>added stop on error functionality<br>default for inputs IN is now TRUE<br>renamed variable state to step                                                                    |
| 14 | SH_2          | 1.1 | 20.01.07 | Added input N to specify the amount of samples for calculations<br>added output trig                                                                                                                                                |
| 14 | DWORD_TO_STRF | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | _ARRAY_INIT   | 1.1 | 06.01.07 | Changed type of function to BOOL<br>function now returns TRUE                                                                                                                                                                       |
| 14 | SUN_POS       | 1.0 | 01.02.07 | New Module                                                                                                                                                                                                                          |
| 14 | SCALE_B4      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                                         |
| 14 | SCALE_X8      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | STAIR2        | 1.0 | 28.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | HYST_3        | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | GEN_SIN       | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | SIGN_R        | 1.1 | 19.01.07 | changed method of function for better compatibility to other systems                                                                                                                                                                |
| 14 | RMP_W         | 1.2 | 22.01.07 | Deleted unused variable step                                                                                                                                                                                                        |
| 14 | KMH_TO_MS     | 1.0 | 04.02.07 | New Module                                                                                                                                                                                                                          |
| 14 | ACTUATOR_2P   | 1.2 | 17.01.07 | Deleted unused variable time_overflow                                                                                                                                                                                               |
| 14 | ACTUATOR_3P   | 1.2 | 23.01.07 | Deleted unused variables „force_on“ and „force_off“                                                                                                                                                                                 |
| 14 | FT_PT2        | 1.1 | 15.01.07 | Improved algorithm for higher accuracy and speed                                                                                                                                                                                    |
| 14 | SENSOR_INT    | 1.1 | 20.01.07 | Deleted input R0 which was not used                                                                                                                                                                                                 |
| 14 | SQRTN         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | BURNER        | 1.3 | 31.12.06 | Added status output to be ESR compatible                                                                                                                                                                                            |
| 14 | F_LIN2        | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_TN16       | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_PID        | 1.2 | 03.01.07 | Added int_band for structural adaptation of PID                                                                                                                                                                                     |
| 14 | BYTE_OF_DWORD | 1.0 | 17.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | AOUT          | 1.1 | 18.01.07 | Renamed module to AOUT<br>changed output to hold 32 bits max<br>corrected an error with sign bit                                                                                                                                    |
| 14 | SUN_TIME      | 1.1 | 20.01.07 | deleted unused variables sun_rineR and sun_setR                                                                                                                                                                                     |
| 14 | SH            | 1.1 | 16.01.07 | Added output trig                                                                                                                                                                                                                   |
| 14 | MAX_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | OM_TO_F       | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | MIN_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_PT1        | 1.2 | 03.01.07 | Corrected an error with falling edge errors<br>added faktor K                                                                                                                                                                       |
| 14 | SEQUENCE_8    | 1.3 | 17.01.07 | Changed output fault to ESR compatible output STATUS<br>added stop on error functionality<br>default for inputs IN is now TRUE<br>renamed variable state to step                                                                    |
| 14 | MULTI_IN      | 1.1 | 01.01.07 | Changed midr to mid3 funktion for better performance                                                                                                                                                                                |
| 14 | STAIR         | 1.0 | 28.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | ACOSH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | BURNER        | 1.2 | 31.12.06 | Added Oil_Temp input                                                                                                                                                                                                                |
| 14 | ACTUATOR_UD   | 1.1 | 22.01.07 | Added status output<br>changed inputs „ton“ and „toff“ to config variables<br>added config var „out_return to enable „yup_in“ and y_dn_in“<br>changed manual mode to override automatic mode                                        |
| 14 | FT_TN8        | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | SCALE_B8      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                                         |
| 14 | DELAY         | 1.2 | 19.01.07 | changed reset to load the value of in instead of 0                                                                                                                                                                                  |
| 14 | CHR           | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | SCALE_X4      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | RDM           | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | LEGIONELLA    | 1.1 | 17.01.07 | Rewritten moule in ST for better performance<br>changed error out to ESR compatible Status output                                                                                                                                   |
| 14 | ACOTH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | WORK_WEEK     | 1.1 | 14.01.07 | deleted unused variable yday                                                                                                                                                                                                        |

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| 14 | SCALE_X2      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DCF_77        | 1.1 | 02.02.07 | change wday and dst outputs when there is no dcf reception                                                                                                                                                  |
| 14 | FT_IMP        | 1.1 | 03.01.07 | Added faktor K                                                                                                                                                                                              |
| 14 | SH_1          | 1.1 | 06.01.07 | Added output trig                                                                                                                                                                                           |
| 14 | MUX_R4        | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DELAY_4       | 1.1 | 19.01.07 | added variable temp to add 1 delay for q1                                                                                                                                                                   |
| 14 | WORD_OF_DWORD | 1.0 | 17.01.07 | New Module                                                                                                                                                                                                  |
| 14 | _ARRAY_SORT   | 1.1 | 06.01.07 | Function now returns TRUE                                                                                                                                                                                   |
| 14 | BURNER        | 1.4 | 22.01.07 | Changed output „fail“ to „status“<br>added output „cycles“<br>corrected an error while pre_heat would be active when an error was present<br>added „coil2“ output and „stage2“ input for dual stage burners |
| 14 | TEMP_PT       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | OFFSET_2      | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_PT1        | 1.1 | 01.01.07 | Corrected error during startup<br>corrected error with very small time values < 10ms                                                                                                                        |
| 14 | COSH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | TANH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_DERIV      | 1.1 | 03.01.07 | Added init code for startup<br>set default for K to 1                                                                                                                                                       |
| 14 | RMP_SOFT      | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ATANH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_INT        | 1.2 | 15.01.07 | Added default value for K = 1                                                                                                                                                                               |
| 14 | COTH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DWORD_OF_BYTE | 1.0 | 24.01.07 | New Module                                                                                                                                                                                                  |
| 14 | MS_TO_KMH     | 1.0 | 04.02.07 | New Module                                                                                                                                                                                                  |
| 14 | RMP_B         | 1.2 | 17.01.07 | Renamed input UD to UP for better usability<br>deleted unused variable step                                                                                                                                 |
| 14 | TEMP_NI       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | ESR_MON_B8    | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ESR_MON_R4    | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_AVG        | 1.1 | 07.01.07 | Changed reset logic to load buffer with IN instead of 0<br>added EN input for better flow control<br>added init to avoid erroneous startup conditions<br>deleted unused variable CNT                        |
| 14 | OFFSET        | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | EXEC          | 1.1 | 06.02.07 | Corrected error with COS function<br>divide by 0 wilol return an error now                                                                                                                                  |
| 14 | AIN           | 1.2 | 19.01.07 | fixed an error with sign bit.                                                                                                                                                                               |
| 14 | LTIME         | 1.2 | 17.01.07 | Added input UTC instead of internal SYSRTCGETTIME for better compatibility with different Hardware<br>corrected an error while DST_ENABLE would not be checked before DST is enabled                        |
| 14 | BURNER        | 1.1 | 31.12.06 | Added Pre heating time and output<br>flame monitor before ignition will generate error output now<br>added output for Kwh                                                                                   |
| 14 | WINDOW        | 1.1 | 22.01.07 | Changed input arrangement                                                                                                                                                                                   |
| 14 | TEMP_SI       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | WORD_OF_BYTE  | 1.0 | 24.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ESR_COLLECT   | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | SINH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_TN64       | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                  |
| 14 | MID_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                  |
| 14 | SCALE_B2      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                 |
| 13 | ACTUATOR_2P   | 1.1 | 07.10.06 | changed error, pos could not reach 1 or 0 under certain conditions due to error in code                                                                                                                     |
| 13 | DATE_ADD      | 1.0 | 27.12.06 | New Module                                                                                                                                                                                                  |
| 13 | TEMP_EXT      | 1.0 | 06.12.06 | New Module                                                                                                                                                                                                  |
| 13 | K_TO_C        | 1.0 | 04.12.06 | New Module                                                                                                                                                                                                  |
| 13 | RMP_B         | 1.1 | 15.10.06 | Added output busy                                                                                                                                                                                           |
| 13 | FIND_NONUM    | 1.0 | 06.10.06 | New Module                                                                                                                                                                                                  |
| 13 | FT_PID        | 1.1 | 01.12.06 | Changed algorithm to „Trapezregel“ for higher accuracy                                                                                                                                                      |
| 13 | TIMER_1       | 1.2 | 25.12.06 | Corrected an error with timer overflow                                                                                                                                                                      |
| 13 | TIMER_1       | 1.1 | 01.12.06 | Corrected an error whith pulse width = 0                                                                                                                                                                    |
| 13 | C_TP_K        | 1.0 | 04.12.06 | New Module                                                                                                                                                                                                  |
| 13 | CYCLE_TIME    | 1.1 | 12.12.06 | Added output cycles                                                                                                                                                                                         |
| 13 | TRIM          | 1.0 | 06.10.06 | New Module                                                                                                                                                                                                  |
| 13 | FINDB_NUM     | 1.0 | 06.10.06 | New Module                                                                                                                                                                                                  |
| 13 | MUX_4         | 1.1 | 05.10.06 | rewritten moule in ST for better performance                                                                                                                                                                |

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|    |                |     |          |                                                                                                       |
|----|----------------|-----|----------|-------------------------------------------------------------------------------------------------------|
| 13 | FT_RMP         | 1.1 | 08.10.06 | Added UD output                                                                                       |
| 13 | FIND_NUM       | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | ACTUATOR_2P    | 1.1 | 07.10.06 | changed „on“ and „off“ to „force“ and „force_in“ for better usability.                                |
| 13 | SEQUENCE_4     | 1.2 | 01.12.06 | corrected an error in sequence logic<br>added init at startup to prevent erroneous startup conditions |
| 13 | MUX_2          | 1.1 | 05.10.06 | rewritten moule in ST for better performance                                                          |
| 13 | UPPERCASE      | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | DAYS_DELTA     | 1.0 | 27.12.06 | New Module                                                                                            |
| 13 | FT_INT         | 1.1 | 13.12.06 | Changed algorithm to „Trapezregel“ for higher accuracy                                                |
| 13 | ACTUATOR_3P    | 1.1 | 19.10.06 | Added code to avoid overrun of the end_switch                                                         |
| 13 | OSCAT_VERSION  | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | LOWERCASE      | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | FINDB          | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | REAL_TO_STRF   | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | ACTUATOR_PUMP  | 1.1 | 27.12.06 | Fixed an error while the pump would run for t_min at startup                                          |
| 13 | EASTER         | 1.0 | 27.12.06 | New Module                                                                                            |
| 13 | DIFFER         | 1.0 | 08.10.06 | New Module                                                                                            |
| 13 | FINDB_NONUM    | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | SEQUENCE_8     | 1.2 | 01.12.06 | corrected an error in sequence logic<br>added init at startup to prevent erroneous startup conditions |
| 13 | STATUS_TO_ESR  | 1.0 | 06.10.06 | New Module                                                                                            |
| 12 | SECOND         | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | RMP_W          | 1.1 | 04.10.06 | added the busy output which signals that the ramp is running.                                         |
| 12 | ARRAY_SUM      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_TREND    | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | LTIME          | 1.1 | 02.10.06 | corrected an error where dst would be delayed by 0.1second                                            |
| 12 | LEAP_OF_DATE   | 1.1 | 02.10.06 | Corrected an error while year 1970 would not work                                                     |
| 12 | CHECK_PARITY   | 1.0 | 01.10.06 | New Module                                                                                            |
| 12 | DELAY          | 1.1 | 01.10.06 | Corrected an error in buffer management                                                               |
| 12 | ARRAY_MIN      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_MAX      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | SEQUENCE_4     | 1.1 | 01.10.06 | Added output any<br>corrected delay to occur aftzer event and not before                              |
| 12 | LEAP_YEAR      | 1.1 | 02.10.06 | Corrected an error while year 1970 would not work                                                     |
| 12 | ARRAY_AVG      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_SPR      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | LOCAL_TIME     | 1.1 | 02.10.06 | Corrected an error while DST would be calculated wrong                                                |
| 12 | HOUR           | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | MULTIME        | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | EVEN           | 1.0 | 01.10.06 | New Module                                                                                            |
| 12 | DEW_TEMP       | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | MINUTE         | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | SEQUENCE_8     | 1.1 | 01.10.06 | Added output any<br>corrected delay to occur aftzer event and not before                              |
| 11 | SH_2           | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | SH_T           | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | FT_MIN_MAX     | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | PARITY         | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_LIN          | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_QUAD         | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | AIN            | 1.1 | 18.08.06 | fixed an error with low value negative and high value 0.                                              |
| 11 | ROUND          | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_POWER        | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | WEEKDAY        | 1.1 | 21.08.06 | Coorected an error in formula                                                                         |
| 10 | TP_R           | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | SET_DT         | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | M_TX           | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | RAD            | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | INC_DEC        | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | MINUTE_TO_TIME | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | SHR_4UDE       | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | DAY_OF_YEAR    | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | HYST_2         | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | MUX_4          | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | SECOND_TO_TIME | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | FF_D2E         | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | CLK_4MS        | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | DST            | 1.0 | 04.08.06 | New Module                                                                                            |

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|    |              |     |          |            |
|----|--------------|-----|----------|------------|
| 10 | FRACT        | 1.0 | 04.08.06 | New Module |
| 10 | SET_TOD      | 1.0 | 04.08.06 | New Module |
| 10 | FF_D4E       | 1.0 | 04.08.06 | New Module |
| 10 | CLK_DIV      | 1.0 | 04.08.06 | New Module |
| 10 | CLK_2MS      | 1.0 | 04.08.06 | New Module |
| 10 | PT_TO_F      | 1.0 | 04.08.06 | New Module |
| 10 | CLK_PULSE    | 1.0 | 04.08.06 | New Module |
| 10 | M_T          | 1.0 | 04.08.06 | New Module |
| 10 | DAY_OF_MONTH | 1.0 | 04.08.06 | New Module |
| 10 | DAY_TO_TIME  | 1.0 | 04.08.06 | New Module |
| 10 | RES_PT       | 1.1 | 04.08.06 | New Module |
| 10 | FIFO_16      | 1.0 | 04.08.06 | New Module |
| 10 | STACK_32     | 1.0 | 04.08.06 | New Module |
| 10 | F_TO_PT      | 1.0 | 04.08.06 | New Module |
| 10 | RES_NI       | 1.0 | 04.08.06 | New Module |
| 10 | CLK_1MS      | 1.0 | 04.08.06 | New Module |
| 10 | MUX_2        | 1.0 | 04.08.06 | New Module |
| 10 | BIT_OF_DWORD | 1.0 | 04.08.06 | New Module |
| 10 | FF_JKE       | 1.0 | 04.08.06 | New Module |
| 10 | LTCH_4       | 1.0 | 04.08.06 | New Module |
| 10 | GEN_SQ       | 1.0 | 04.08.06 | New Module |
| 10 | GEN_BIT      | 1.0 | 04.08.06 | New Module |
| 10 | SHR_8UDE     | 1.0 | 04.08.06 | New Module |
| 10 | HOUR         | 1.0 | 04.08.06 | New Module |
| 10 | SHR_4E       | 1.0 | 04.08.06 | New Module |
| 10 | B_TRIG       | 1.0 | 04.08.06 | New Module |
| 10 | BYTE_OF_BIT  | 1.0 | 04.08.06 | New Module |
| 10 | BYTE_TO_BIT  | 1.0 | 04.08.06 | New Module |
| 10 | SHR_8PLE     | 1.0 | 04.08.06 | New Module |
| 10 | C_TO_F       | 1.0 | 04.08.06 | New Module |
| 10 | FF_DRE       | 1.0 | 04.08.06 | New Module |
| 10 | STACK_16     | 1.0 | 04.08.06 | New Module |
| 10 | SET_DATE     | 1.0 | 04.08.06 | New Module |
| 10 | DEG          | 1.0 | 04.08.06 | New Module |
| 10 | SWITCH_I     | 1.0 | 04.08.06 | New Module |
| 10 | HOUR_TO_TOD  | 1.0 | 04.08.06 | New Module |
| 10 | YEAR         | 1.0 | 04.08.06 | New Module |
| 10 | HYST_1       | 1.0 | 04.08.06 | New Module |
| 10 | MONTH        | 1.0 | 04.08.06 | New Module |
| 10 | HOUR_TO_TIME | 1.0 | 04.08.06 | New Module |
| 10 | LTCH         | 1.0 | 04.08.06 | New Module |
| 10 | CLK_8MS      | 1.0 | 04.08.06 | New Module |
| 10 | RES_SI       | 1.0 | 04.08.06 | New Module |
| 10 | F_TO_C       | 1.0 | 04.08.06 | New Module |
| 10 | FIFO_32      | 1.0 | 04.08.06 | New Module |